

BEYOND LOCAL LYAPUNOV STABILITY: INVERSE MOMENTS AND NONLINEAR RETURN IN SGD

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Working manuscript

ABSTRACT

Stochastic gradient descent can remain confined while local linear models predict expansion. We separate the return mechanism creating a stationary distribution from the moments describing its central shape. Scalar quadratic SGD gives $a_B = 1 - \eta h_B$: positive moments govern perturbation growth, while a negative root $\mathbb{E}|a_B|^{-\alpha} = 1$ describes small stationary amplitudes in a returned process. The harmonic reference $\mathbb{E}|a_B|^{-1} = 1$ separates central concentration from depletion under explicit assumptions. A matrix renewal formulation retains direction dynamics. Coercive nonconvex SGD constructions then establish attracting invariant laws beyond local expansion, including a control with a positive stationary tangent exponent. Neural batch-size–step-size heatmaps and fixed-frame continuations show distinct scalar regimes and alternating two-lobe occupancy while transported tangents grow and loss decreases. These results identify what can coexist beyond a local edge and why neural bimodality needs direct, time-resolved measurement.

1 INTRODUCTION

A neural network can keep learning while local curvature predicts expansion. Can the optimizer remain statistically confined, and what shape does its occupation distribution have? An expanding deterministic quadratic has no return mechanism. Random curvature further separates the boundaries for mean-square, logarithmic, and inverse-moment growth.

A scalar quadratic gives the multiplier $a_B = 1 - \eta h_B$, with sampled curvature h_B and step size η . Its moments $m(p) = \mathbb{E}|a_B|^p$ separate several questions. Positive orders describe perturbation growth; the logarithmic average describes a typical product. The negative moment $m(-1)$ measures concentration toward zero. We call $H = 1/m(-1)$ the *harmonic gain*; its central-shape interpretation requires an explicit return model.

Classical Kesten–Goldie theory describes large values of a contractive random affine recursion. Our central-mass formulation combines local expansion with outer return: a *negative* moment root then controls small stationary amplitudes. We use reflected-walk and renewal ideas, retain the evolving direction in a matrix formulation, and exhibit nonconvex SGD losses whose gradients supply return.

We separate local growth, nonlinear recurrence, and coordinate-distribution shape through three results:

1. A scalar-to-matrix central renewal formulation with explicit return assumptions. It identifies the inverse root and explains when the harmonic reference separates central concentration from depletion.
2. Concrete coercive-loss SGD constructions proving that nonlinear return permits an attracting invariant law beyond local expansion. A separate piecewise-smooth construction also has a provably positive stationary tangent exponent.
3. Neural heatmaps and fixed-frame continuations revealing a concrete diagnostic disagreement: an MLP has negative late fixed-axis log gain, positive transported growth, and alternating two-lobe occupancy under iid batches. A short CNN candidate fails confirmation.

The constructions prove what can coexist; the neural measurements test these distinctions over finite training windows.

2 SCALAR QUADRATICS AND THE NEGATIVE-MOMENT BOUNDARY

2.1 FROM ONE DETERMINISTIC EDGE TO SEVERAL STOCHASTIC DIAGNOSTICS

For deterministic $L(x) = hx^2/2$ with $h > 0$, the entire orbit is

$$x_t = (1 - \eta h)^t x_0. \quad (1)$$

The condition $0 < \eta h < 2$ contracts every nonzero start to zero. At $\eta h = 2$ the sign alternates without decay; above two the magnitude diverges. A fixed quadratic therefore cannot explain bounded large-step fluctuations after its expanding boundary. Random curvature changes the product law, and nonlinear curvature can change the orbit's return.

On the scalar sampled quadratic $\ell_B(x) = h_B x^2/2 - \xi_B x$, SGD is exactly

$$X_{t+1} = A_{t+1} X_t + \eta \xi_{t+1}, \quad A_B = 1 - \eta h_B. \quad (2)$$

Two trajectories using the same sampled batches differ by $\Delta_{t+1} = A_{t+1} \Delta_t$. Write $M = |A_B|$, so moments always use the absolute multiplier, and define

$$m(p) = \mathbb{E} M^p, \quad \gamma = \mathbb{E} \log M, \quad H = \frac{1}{m(-1)}. \quad (3)$$

Here H is the harmonic gain; its inverse $m(-1)$ is the negative-moment coefficient. We only assign a finite inverse-moment estimate this meaning when the population inverse moment exists. For constant curvature and $p \neq 0$, $m(p)^{1/p} = M$, $\gamma = \log M$, and $H = M$: the boundaries all coincide at $|1 - \eta h| = 1$. Curvature randomness separates them.

Proposition 1 (Scalar moment ordering). *Let $M > 0$, $\mathbb{E} |\log M| < \infty$, and suppose the displayed moments are finite. Then*

$$\log H \leq \gamma \leq \log m(1) \leq \frac{1}{2} \log m(2). \quad (4)$$

For iid multipliers and $\Delta_0 \neq 0$, $t^{-1} \log |\Delta_t/\Delta_0| \rightarrow \gamma$ almost surely and $\mathbb{E} |\Delta_t/\Delta_0|^p = m(p)^t$. Consequently $H > 1$ implies $\gamma > 0$. The converse need not hold.

The use of absolute values matters: $m(1)$ is a mean-absolute multiplier, not the signed mean $\mathbb{E} a_B$. For $p > 0$, $m(p) < 1$ means contraction of a p th perturbation moment. For $p < 0$ it means contraction of an *inverse distance*, so it measures repulsion from zero, rather than contraction toward zero. The logarithmic criterion lies between these orders. We use the term harmonic expansion for $H > 1$ while keeping this direction of the inequality explicit.

A concrete two-curvature law already separates the references. Let $a_B = -1/2$ with probability u and $a_B = -2$ otherwise; both are realizable with positive quadratic curvatures. Then

$$\gamma = (1 - 2u) \log 2, \quad H = (1/2 + 3u/2)^{-1}. \quad (5)$$

For $1/3 < u < 1/2$, the product is logarithmically expanding but $H < 1$. For $u < 1/3$, both diagnostics expand. No dimension or gradient-alignment assumption is needed for this separation.

The classical Kesten–Goldie result concerns the *affine* recursion (2): with $\gamma < 0$, nondegenerate additive forcing, and the usual renewal assumptions, its stationary upper tail has exponent $\kappa > 0$ solving $m(\kappa) = 1$ (Kesten, 1973; Goldie, 1991). This is a baseline, not a theorem of post-expansion confinement. In particular, the nondegenerate Gaussian affine model used in Appendix A has no invariant probability when $\gamma > 0$. An additional return mechanism is needed.

2.2 NONLINEAR RETURN CHANGES AN UPPER-TAIL QUESTION INTO A CENTRAL-MASS QUESTION

An *invariant law* π satisfies $X_0 \sim \pi \Rightarrow X_t \sim \pi$ at every time. It is *attracting* when other initial laws converge to it in distribution. These statements concern the law of a state; they need not imply contraction of paths using the same random updates.

The simplest exact return model is saturated amplitude:

$$R_{t+1} = \min\{r_*, M_{t+1} R_t\}, \quad 0 < R_t \leq r_*. \quad (6)$$

Here $r_* > 0$ is the outer return radius. The model separates local multiplicative motion from return at this scale. It is an explicit model, not an identity for a globally quadratic loss.

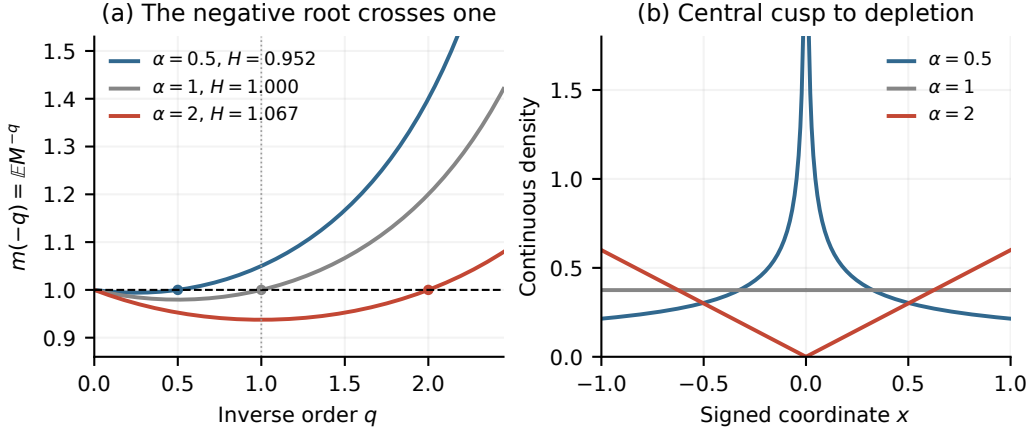


Figure 1: **An exact central-shape law for a returned multiplier.** Let $M = e^{V-U}$ with independent $U \sim \text{Exp}(b)$, $V \sim \text{Exp}(3)$, and $b = 3 + \alpha$. The saturated process $R^+ = \min\{1, MR\}$ has positive local log drift in all three cases. (a) Its inverse root $m(-\alpha) = 1$ crosses order one as H crosses one. (b) The continuous part of the symmetrically signed invariant law is $(3/b)(\alpha/2)|x|^{\alpha-1}$ on $0 < |x| < 1$: a cusp, a flat center, or central depletion. Each endpoint also has an atom of mass $(1 - 3/b)/2$, which is not drawn as a density. These are analytic curves; Proposition 7 in Appendix A verifies the law exactly.

Theorem 2 (An inverse-moment law after local expansion). *Suppose $\mathbb{E}|\log M| < \infty$, $\gamma > 0$, $\log M$ is nonarithmetic, and $m(-\alpha) = 1$ for some $\alpha > 0$. Assume $m(-q)$ is finite in a neighborhood of α and $0 < -\mathbb{E}[M^{-\alpha} \log M] < \infty$. Then (6) has a unique invariant probability on $(0, r_*]$, and for some $C \in (0, \infty)$,*

$$\mathbb{P}_\pi(R \leq r) \sim C(r/r_*)^\alpha, \quad r \downarrow 0. \quad (7)$$

If, additionally, $\log M$ has a bounded continuous density of compact support, then $p_R(r) \sim C\alpha r_^{-\alpha} r^{\alpha-1}$. When the negative-moment curve is strictly convex and finite through 1 and α ,*

$$H < 1 \iff \alpha < 1, \quad H = 1 \iff \alpha = 1, \quad H > 1 \iff \alpha > 1. \quad (8)$$

Thus positive local log growth can coexist with a central density cusp ($0 < \alpha < 1$); the harmonic crossing separates this cusp from central depletion ($\alpha > 1$) in this return model. For a reflection-symmetric signed coordinate, the same power occurs on each side. Central depletion is a local shape conclusion. Exactly two modes additionally requires control of the outer density, and arbitrary reinjection can reverse the conclusion (Appendix A.7).

Proof idea. The change of variable $W = \log(r_*/R)$ gives $W_{t+1} = \max\{0, W_t - \log M_{t+1}\}$. This is a reflected random walk with negative mean increment. Its invariant law is the supremum of the unreflected walk. The nonarithmetic assumption rules out log-radius lattice effects. Tilting the increment law by $M^{-\alpha}$ reverses its drift; the overshoot renewal theorem gives an exponential tail in W , hence the power (7) in R . The root therefore has the opposite sign from the Kesten upper-tail root.

3 A MATRIX RENEWAL FORMULATION

3.1 RANDOM MATRICES: RADIUS AND DIRECTION MUST EVOLVE TOGETHER

For a quadratic in d parameters, the linear update is $U_{t+1} = A_{t+1}U_t$, where $A = I - \eta H_B$. Writing $U = rs$, with $\|s\| = 1$, gives

$$r^+ = r\|As\|, \quad s^+ = As/\|As\|. \quad (9)$$

Thus a matrix changes both the amplitude and the direction of the next update. Its log gains need not be independent. The appropriate analogue of the scalar moment $\mathbb{E}|a|^q$ is the growth rate of the

positive operator

$$(P_q f)(s) = \mathbb{E} \left[\|As\|^q f \left(\frac{As}{\|As\|} \right) \right], \quad k(q) = \text{spr}(P_q). \quad (10)$$

Here spr denotes spectral radius: $k(q)$ is the long-run moment-growth factor, including changes of direction. The operator weights each next-direction value of f by the current radial gain to order q . When the original angular chain is ergodic, identifies the top product exponent, and the pressure is differentiable at zero, $\gamma = (\log k)'(0)$ is the linear-product Lyapunov exponent. In one dimension, $k(q) = \mathbb{E}|a|^q$. Classical matrix renewal theory uses a positive root to describe large values of a contractive affine recursion (Kesten, 1973; Guivarc’h & Le Page, 2016). The following result uses a negative root to describe small amplitudes of a locally expanding process supplied with return. Its proof changes the sampling law (a probability *tilt*) to emphasize rare inward excursions. The assumptions ensure that directions mix and that this tilted radius drifts inward; Appendix B gives the formulas and an explicit noncommuting example.

Theorem 3 (Matrix central law with finite return cycles). *Suppose the linear product is locally expanding, $\gamma > 0$, and $k(-\alpha) = 1$ for some $\alpha > 0$. Assume the negative-order spectral and Markov renewal conditions in Assumption 1, including a bounded strictly positive eigenfunction and nonarithmetic angular mixing. Start an exact linear excursion below radius r_0 and restart it from the same independent entry law on its first exit above r_0 . If the mean excursion length is finite and the entry law has a finite inverse α -moment, then its invariant law satisfies*

$$\lim_{r \downarrow 0} \frac{\pi(\|U\| \leq r, U/\|U\| \in D)}{(r/r_0)^\alpha} = C\chi(D), \quad (11)$$

for angular continuity sets D , provided the tilted excursion has a positive probability of never returning to the upper boundary. Here $C > 0$ and χ is a probability measure on directions. The matrices may be noncommuting.

What this establishes. Finite return cycles create stationarity; the negative moment root determines how often stationary excursions reach the center. The proof tilts the joint radius–direction process, applies a Markov renewal theorem, and subtracts occupation after upper exit. It does not replace matrix products by Rayleigh quotients. For a genuine nonlinear SGD chain, Proposition 9 gives a precise transfer criterion in terms of its entry law and local transition kernel. Those conditions are additional to positive local growth and are not established for our neural networks.

Which density can lose its central peak? With an additional renewal-density regularity assumption, (11) gives radial density $p_R(r) \asymp r^{\alpha-1}$. If a joint local density limit in radius and direction also holds, dividing by the polar volume factor gives ambient density $p_U(rs) \asymp r^{\alpha-d}$ along directions with positive limiting density. For a strictly convex log pressure with roots $-\alpha$ and 0, $k(-1) < 1$ means $\alpha > 1$, whereas $k(-d) < 1$ means $\alpha > d$. A fixed coordinate integrates over directions and has no such universal threshold: Proposition 10 gives a stationary nonlinear SGD example with both inequalities but centrally unimodal fixed-coordinate marginals. This distinction motivates direct fixed-direction measurements in the neural experiments.

4 NONCONVEX SGD SUPPLIES ITS OWN RETURN

The next construction makes return part of the sampled loss. Here the local exponent freezes a Jacobian at the population minimizer; the occupied exponent follows Jacobians along a stationary trajectory.

To prove that return can arise from a gradient update, rather than from an imposed cap, draw $h \in \{.95, 1.05\}$ equiprobably and $e \sim N(0, \sigma^2)$, independently, with $\sigma > 0$. Use

$$\ell(x; h, e) = \frac{1}{2}(\sqrt{h}x - e)^2 - \frac{.92}{2}\{x^2 - \log(1 + x^2)\}. \quad (12)$$

Every sampled loss is coercive: it tends to infinity as $|x| \rightarrow \infty$. The population loss, up to a constant, is $L(x) = .04x^2 + .46 \log(1 + x^2)$, with unique minimizer zero but negative curvature on part of the line.

Theorem 4 (Smooth nonconvex SGD can return despite central expansion). *More generally, replace .92 in (12) by β and draw $h = \beta + \mu \pm s$ equiprobably, with $0 < s < \mu$ and $\beta > 8\mu$. At batch size one, every step $0 < \eta < 2/(\mu + s)$ has a unique attracting invariant probability with a smooth, strictly positive, even density and all finite polynomial moments. Its central Jacobian is $1 - \eta h$, whereas its outer slopes are $1 - \eta(\mu \pm s)$.*

For $(\beta, \mu, s, \eta) = (.92, .08, .05, 2.015)$, the central absolute gains are .91425, 1.11575, and

$$\gamma_0 = 0.009938 \dots > 0, \quad H_0 = 1.004999 \dots > 1, \quad \alpha_0 = 2.017427 \dots > 1, \quad (13)$$

where $\frac{1}{2}(.91425^{-\alpha_0} + 1.11575^{-\alpha_0}) = 1$. The outer slopes are .93955, .73805, both strictly contracting.

Proof idea. The exact update is $X^+ = A_\infty X - \eta\beta X/(1 + X^2) + \eta\sqrt{h}e$, with $A_\infty = 1 - \eta(h - \beta)$. The middle term is bounded and the outer slopes contract. For every $p > 0$, the bound $\mathbb{E}[|X^+|^p | X = x] \leq c_p|x|^p + C_p$, with $c_p < 1$, controls excursions to infinity. This inward drift and the positive Gaussian transition density give recurrence and uniqueness. At the displayed step, the local random product has positive mean log gain because curvature is much larger near the origin. This theorem proves recurrence after the *central* Lyapunov and harmonic boundaries; it does not identify the tangent exponent along the invariant trajectory or infer its modes. These are separate empirical measurements below.

For this example α_0 describes the *frozen* multiplier law. At every fixed $\sigma > 0$, the actual invariant density is strictly positive at zero, so a zero-noise power-law depletion cannot be inferred literally as $x \rightarrow 0$. The recurrence theorem and the central renewal theorem establish complementary statements, with different hypotheses.

4.1 WHAT BALANCES AT EQUILIBRIUM?

For plain SGD $X^+ = X - \eta g_B(X)$, stationarity with finite second moments gives the exact identity

$$2\mathbb{E}_\pi \langle X, \nabla L(X) \rangle = \eta \mathbb{E}_{\pi, B} \|g_B(X)\|^2. \quad (14)$$

Indeed, expand $\|X^+\|^2 - \|X\|^2$, condition on X , and integrate against π . The identity is justified for Theorem 4, whose loss gradients have linear growth and whose invariant law has all moments. Inward motion on the left balances the squared random updates on the right. Neither side is determined by the Jacobian at zero. The equilibrium is a distribution over visited states, rather than a point where the frozen multiplier must equal one.

The balance is visible directly in the conditional radial drift of the smooth example. At batch one, define $d(x) = \mathbb{E}[(X^+)^2 - X^2 | X = x]$. Independence and the centered residual give

$$d(x) = \left[\left(1 - \eta\mu - \frac{\eta\beta}{1 + x^2} \right)^2 + \eta^2 s^2 - 1 \right] x^2 + \eta^2 \sigma^2 (\beta + \mu). \quad (15)$$

For the displayed parameters, small states have outward drift, whereas $d(x)$ is negative at sufficiently large $|x|$. The invariant law averages these regions so that $\mathbb{E}_\pi d(X) = 0$. A zero of $d(x)$ is a local balance radius, not an assertion that the distribution is a point mass there.

In the smooth construction, the outer slopes control return while the central sampled curvatures control frozen expansion. A separate exact control shows that the *occupied* tangent exponent can be positive as well. Let $a \in \{.58, .62\}$ equiprobably, $\xi \sim \text{Unif}[-.38, .38]$, and set

$$\ell(x; a, \xi) = x^2/2 - a\Psi(x) - \xi x, \quad \Psi(x) = \begin{cases} x - x|x|, & |x| \leq 1, \\ -x + \text{sign}(x), & |x| > 1. \end{cases} \quad (16)$$

These coercive losses are C^1 with Lipschitz gradients and finitely many Hessian kinks. Their population loss is nonconvex.

Theorem 5 (Stationarity with positive occupied growth). *At $\eta = 1$, SGD on (16) has a unique attracting invariant law supported on $[-1, 1]$, all moments finite, and stationary common-noise tangent exponent*

$$\lambda_\pi = \frac{1}{2} \log(1.16 \cdot 1.24) = 0.18176569 \dots > 0. \quad (17)$$

For the same loss, all $\eta \in [.98, 1]$ have a unique attracting invariant law and $\lambda_\pi \geq \log(1.1168) > 0$.

At $\eta = 1$, the update is the randomly perturbed tent map $X^+ = a(1 - 2 \min\{|X|, 1\}) + \xi$. A two-step minorization gives uniform mixing, while its derivative magnitude on the occupied interval is exactly $2a > 1$, also giving an occupied harmonic gain above one. Appendix D proves both assertions and the step-size interval. This is an explicit SGD realization of a familiar expanding-map mechanism. It shows that attraction of *distributions* and growth of infinitesimal common-noise perturbations can coexist; no mode-count claim follows from the positive exponent.

5 FROM THE THEORY TO MEASURABLE NEURAL QUANTITIES

The scalar reduction is exact in one dimension. In a network, it is a diagnostic of a specified frame. At state θ , let $J_B = I - \eta H_B(\theta)$ be the plain-SGD Jacobian and let q be a unit vector. Its scalar compression and reset norm are

$$a_q = q^\top J_B q, \quad \|J_B q\|^2 = a_q^2 + \|(I - qq^\top)J_B q\|^2. \quad (18)$$

The second term is the motion discarded by scalar projection. Reinitializing q after every step also discards the direction inherited from earlier steps. The matrix-product measurement instead uses

$$v_{t+1} = J_{B_t}(\theta_t)v_t, \quad \hat{\lambda}_{[s, s+T)} = \frac{1}{T} \sum_{t=s}^{s+T-1} \log \frac{\|J_{B_t}(\theta_t)v_t\|}{\|v_t\|}. \quad (19)$$

Numerical normalization preserves v_t 's direction and accumulates the logarithms. This is a finite-time transported growth rate, becoming a Lyapunov exponent only under a justified long-time limit. Even one-step norms do not determine this limit: alternating $\text{diag}(2, .1)$ and $\text{diag}(.1, 2)$ have norm two at each step but two-step product $.2I$.

For the batch-gradient axis $q_B = g_B/\|g_B\|$, with $g_B = \nabla L_B(\theta)$, we write $a_g = 1 - \eta s_B$, where $s_B = q_B^\top H_B q_B$. This uses the same batch for curvature and direction. We report $\hat{\Lambda}_g = \text{mean log } |a_g|$ and $\hat{H}_g = (\text{mean } |a_g|^{-1})^{-1}$. These moments of a random compression do not equal moments of a batch-averaged sharpness, and q_B is neither fixed nor transported. A finite harmonic estimate also cannot certify that its population inverse moment is finite: rare near-zero gains can dominate it.

Finally, a projected parameter histogram must use one fixed axis and one fixed center. Otherwise the coordinate itself changes with the optimizer. We therefore pair dense fixed-frame traces with independent conditional probes and the derivative product in (19). Small windows reveal transient structure; bandwidth changes, ordered traces, and independent batch streams help distinguish a recurring pair of lobes from drift or a fragile histogram split.

6 NEURAL OBSERVATIONS ACROSS REGIMES AND WINDOWS

Neural diagnostics separate the two scalar edges. We study a three-convolution SiLU CNN and a two-hidden-layer, width-512 SiLU MLP, trained by plain SGD on the same 8,192-example CIFAR-10 subset with squared loss. At a fixed state, write $g_B = \nabla L_B(\theta)$, $s_B = g_B^\top \nabla^2 L_B(\theta) g_B / \|g_B\|^2$, and $a_g = 1 - \eta s_B$. Figure 2 estimates the log and negative-moment diagnostics of this scalar compression across batch sizes and step sizes. For the CNN, $(\eta, B) = (0.16, 128)$ gives $(\hat{\Lambda}_g, \hat{H}_g) = (0.179, 0.760)$, whereas $(0.08, 512)$ gives $(0.176, 1.176)$: similar positive log gain can occur on either side of the harmonic reference. Other settings approach one or both references; for example the MLP at $(0.02, 4096)$ has $(0.0027, 1.0008)$. The settings and checkpoint windows matter: the sweep averages do not assign an architecture a universal stability threshold.

A fixed coordinate reveals short-window structure. We continue a retained MLP checkpoint for 4,096 updates at $B = 4096$, $\eta = 0.04$. Its initial 128-draw conditional probe has $(\hat{\Lambda}_g, \hat{H}_g) = (0.136, 1.144)$; this selected checkpoint differs from the sweep's window averages. We freeze the reference leading-Hessian Ritz vector q_* and record $q_*^\top(\theta_t - \theta_*)$ before every update. The iid-minibatch control retains two-lobed occupancy (Figure 3); the displayed 512-update interval has lag-one correlation -0.985 , consistent with approximate period-two alternation. Meanwhile, continuously propagated tangent vectors grow at approximately 0.00642 per update over the last 2,048 updates, while training loss decreases. Fixed-axis compression has a different late log average,

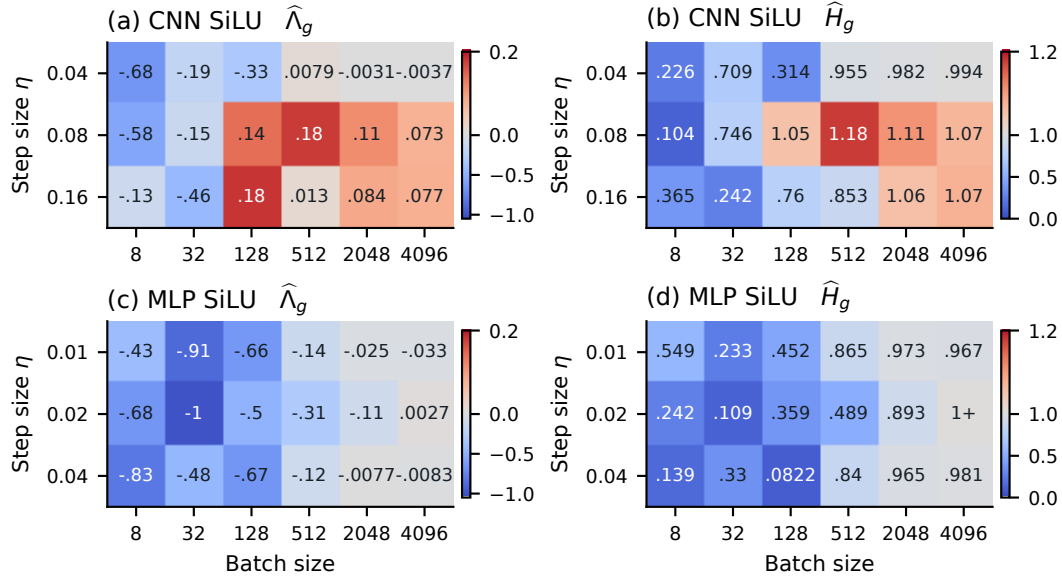


Figure 2: **Neural networks show distinct scalar diagnostic regimes.** Plain SGD on 8,192 CIFAR-10 examples with squared loss. For each run, the last three available conditional probe sets are pooled to estimate $\hat{\Lambda}_g = \text{mean log } |a_g|$ and $\hat{H}_g = (\text{mean } |a_g|^{-1})^{-1}$; cells average these run-level statistics. Color centers are 0 and 1; axes use the actual training step size and batch size. CNN settings include $\hat{\Lambda}_g > 0$ on both sides of $\hat{H}_g = 1$. These are checkpoint-window diagnostics, with varying horizons, not equilibrium or full-network stability certificates. A trailing + preserves a value just above the displayed reference after rounding.

−0.268, because it discards coupling to other directions. Adjacent occupied states can also lie on opposite sides of $\hat{H}_{q_*} = 1$. Thus the observations connect local expansion to bounded finite-time occupancy, but do not establish stationary bimodality, an asymptotic Lyapunov exponent, or a harmonic-threshold cause of neural bimodality. The CNN’s discovery-selected 64-update candidate does not survive the same bandwidth screen in a separate continuation; the complete scans and this negative control are retained in the appendix.

Reading an architecture’s operating regime. The two diagnostics in a cell use the same gain pool and scalar definition; comparisons across cells also change the trained states and checkpoint windows. The MLP at $\eta = .02$, $B = 4096$ approaches both references, whereas the CNN at $\eta = .16$, $B = 128$ is logarithmically expanding with H distinctly below one. The appendix adds ten further vision architectures: some approach the logarithmic reference, some the harmonic reference, and several remain contracting throughout their measured grid. This is a distribution of observed regimes across settings; there is no evidence here that architecture alone determines one preferred edge.

The chosen MLP checkpoint makes an additional distinction especially clear. Its initial batch-gradient law lies beyond both references, but its late fixed-leading-axis log average is negative even while transported tangents grow. The fixed axis retains the geometric identity of the two lobes; the tangent calculation retains the transported direction and its coupling to the other coordinates. Each is useful for a different question. The adjacent-state probes show that even within one oscillation, the fixed-axis harmonic statistic changes substantially. Averaging over the two phases can hide this structure.

Coverage and negative controls. The two MLP samplers both pass the fixed-top-axis shape screen in all windows of lengths 256, 512, and 1,024 in the declared scan. These overlapping windows document persistence across widths, not independent replications. The CNN supplies a useful contrast: its 64-update discovery interval fails the same screen in the confirmation stream, and neither continuation has a passing fixed-top-axis window of length 256 or more. Longer ordinary-network controls in the separate nonconvex campaign also show cutoff escape or parameter drift. Thus declining loss and a positive frozen diagnostic alone do not certify a neural equilibrium.

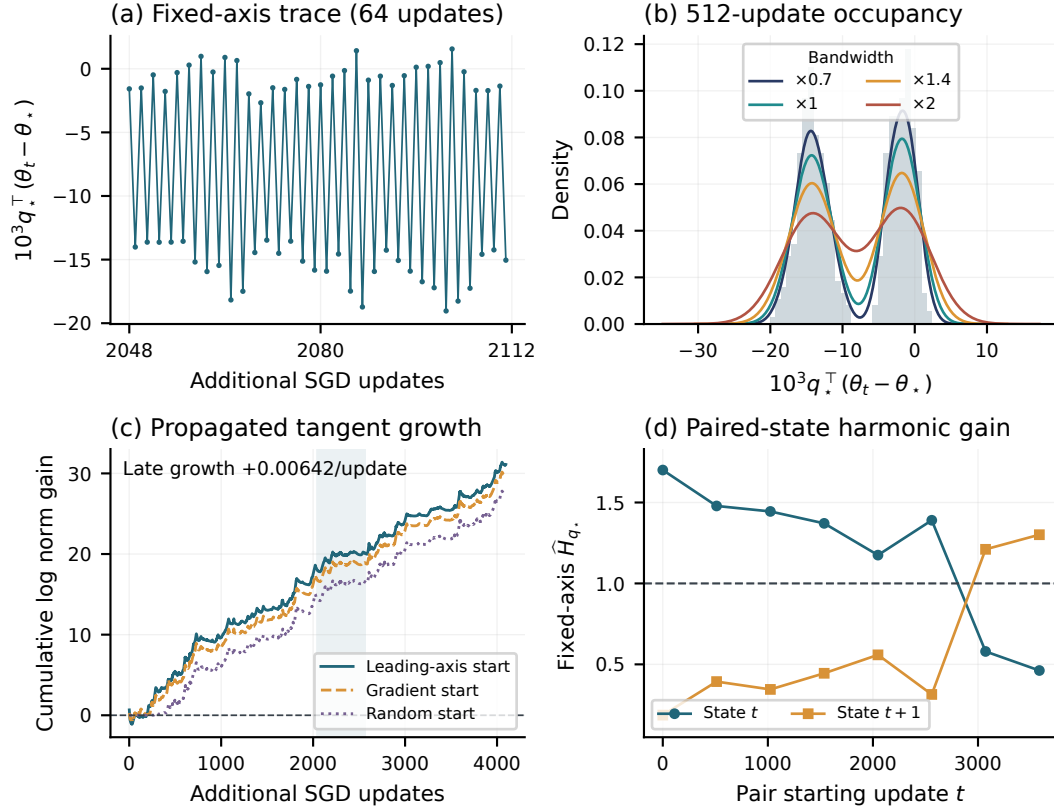


Figure 3: **Local expansion, two-lobed occupancy, and changing phase coexist.** SiLU MLP, CIFAR-10, $B = 4096$, $\eta = 0.04$, with independently sampled minibatches. The reference checkpoint θ_* and its leading Hessian Ritz direction q_* are fixed; this coordinate origin need not be a density center. (a,b) A 64-update trace and its enclosing 512-update occupancy window show alternating lobes across four KDE bandwidths. (c) Continuously transported tangent vectors have late growth $+0.00642$ per update; loss decreases from 0.02485 to 0.00935. (d) The same 24 fresh minibatches probe each adjacent state pair; their fixed-axis harmonic gains lie on opposite sides of one. This is finite-window, approximately period-two occupancy; it does not establish a stationary bimodal law or a universal harmonic transition.

The supplementary atlas preserves missing probe cells and recorded cutoffs. It also includes a 168M-parameter GPT on FineWeb-Edu with token-aligned checkpoint pools. That model uses Adam, so its raw and frozen-preconditioner curvature views are explicitly auxiliary diagnostics; they do not measure the full optimizer-state Jacobian or inherit the plain-SGD theorems.

7 RELATION TO PRIOR WORK AND SCOPE

The edge-of-stability observations of [Cohen et al. \(2021\)](#) motivate studying dynamics near the deterministic quadratic boundary. Batch sharpness and dynamical stochastic stability address different extensions of that picture ([Andreyev & Beneventano, 2025](#); [Chemnitz & Engel, 2025](#)). Local quadratic models can remain predictive over substantial LLM training windows ([Meterez et al., 2026](#)); our distinction concerns what their scalar compressions imply about nonlinear return and occupation shape. Approximate period-two neural traces are also naturally related to two-step dynamics ([Chen & Bruna, 2023](#)).

Our probability tools have substantial classical foundations. Kesten–Goldie and matrix spectral renewal theory concern random recursions ([Kesten, 1973](#); [Goldie, 1991](#); [Guivarc’h & Le Page, 2016](#)); reflected-walk renewal supplies the scalar central exponent, and Markov renewal supplies its directional counterpart ([Alsmeyer, 1997](#)). Nonlinear asymptotically linear systems already have extensive stationary-tail theory ([Alsmeyer, 2016](#); [Alsmeyer et al., 2023](#)). SGD heavy-tail analyses likewise precede this work ([Gürbüzbalaban et al., 2021](#); [Damek & Mentemeier, 2026](#)). We do not claim a new general heavy-tail theorem or first discovery of confined chaotic optimization, which has deterministic and stochastic precedents ([Chen et al., 2023](#); [Tuci et al., 2026](#); [Kalle & Maggioni, 2022](#)).

The central renewal statements here have explicit entry and kernel assumptions. They are proved for the stated return models; their nonlinear transfer conditions have not been verified for ordinary neural SGD. Fixed additive Gaussian noise smooths the center, and matrix radial depletion need not produce two fixed-coordinate modes. The neural examples establish finite-window coexistence and diagnostic disagreement. They do not establish global parameter stationarity, a universal architecture-specific edge, or harmonic expansion as a cause of bimodality.

8 DISCUSSION

Nonlinear return answers why local expansion need not cause escape. Negative moments answer a different question: under a controlled return law, how much stationary mass reaches the center? Matrix direction dynamics then determine which central law is relevant. The experiments show why all three questions matter: a network can lower its loss, display alternating two-lobe occupancy, and have different signs for fixed-axis and transported growth. A useful stochastic-edge description should therefore name its reference state, direction process, and time scale. Establishing a neural equilibrium law and its mode transition requires those measurements together with a verified return mechanism.

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A SCALAR THEORY: ASSUMPTIONS AND COMPLETE DERIVATIONS

A.1 DETERMINISTIC AND IID QUADRATIC PRODUCTS

For $\ell(x) = hx^2/2$, deterministic gradient descent satisfies $x_t = (1 - \eta h)^t x_0$. If $h > 0$, convergence to zero from every finite start occurs precisely when $0 < \eta h < 2$. At $\eta h = 2$, nonzero starts follow a two-cycle rather than converge, and at $\eta h > 2$ their magnitudes diverge. The mean, mean-square, logarithmic, and harmonic multiplier criteria coincide in this deterministic case. This calculation does not describe stationary fluctuations because there is no sampled forcing and no mechanism returning an expanding linear orbit to a bounded scale.

For iid batches, common-noise differences satisfy $\Delta_t/\Delta_0 = \prod_{j=1}^t A_j$, regardless of additive forcing. The logarithm is a sum of iid integrable variables, so the strong law gives the almost-sure exponent in Proposition 1. Independence also gives the exact p -moment identity, for every real p for which the moment is defined. For negative p , this identity describes inverse-distance growth; its contraction means repulsion from zero and must not be called contraction toward zero.

Jensen applied to $\exp(-\log M)$ gives $\log \mathbb{E}M^{-1} \geq -\mathbb{E} \log M$. Jensen applied to $\exp(\log M)$ gives $\mathbb{E} \log M \leq \log \mathbb{E}M$. Cauchy–Schwarz gives $\mathbb{E}M \leq (\mathbb{E}M^2)^{1/2}$. These are precisely (4). If M is nonconstant, the first Jensen inequality is strict whenever both sides are finite, so $H = 1$ then also implies $\gamma > 0$.

For example, let $M = 1/2$ with probability $1/5$ and $M = 2$ with probability $4/5$. Then $\gamma = (3/5) \log 2 > 0$, $m(-1) = 4/5$, $H = 5/4 > 1$, and $m(-2) = 1$. Changing the contracting probability to $2/5$ gives positive $\gamma = (1/5) \log 2$ while $m(-1) = 11/10 > 1$, hence $H < 1$. The separated thresholds therefore occur already for two-point curvature laws, not only for estimated high-dimensional diagnostics.

A.2 THE AFFINE BASELINE AND WHY OUTER RETURN MATTERS

The following restricted setting suffices for an exact comparison and avoids unspoken nondegeneracy exceptions. Let (A_t, N_t) be iid, A_t take finitely many nonzero real values, and, conditionally on A_t , let N_t be centered Gaussian with variance in $[v_-, v_+] \subset (0, \infty)$. Multipliers and noise may be dependent. Consider

$$X_{t+1} = A_{t+1}X_t + r(X_t) + N_{t+1}, \quad \sup_x |r(x)| \leq R < \infty, \quad (20)$$

where r is continuous.

Proposition 6 (Outer logarithmic drift separates recurrence and escape). *If $\mathbb{E} \log |A| < 0$, (20) has a unique positive Harris invariant probability and converges geometrically in a weighted total-variation norm with weight $1 + |x|^p$ for some $0 < p \leq 1$. If $\mathbb{E} \log |A| > 0$, it has no invariant probability. No conclusion at equality is asserted.*

Proof. In the negative case, finiteness and nonvanishing of the multiplier support imply that $q(p) = \mathbb{E}|A|^p$ is differentiable at zero, with $q(0) = 1$ and $q'(0) < 0$. Choose $0 < p \leq 1$ with $q(p) < 1$. Subadditivity gives

$$P(1 + |x|^p) \leq q(p)(1 + |x|^p) + 1 - q(p) + R^p + \mathbb{E}|N|^p.$$

The transition density is a finite mixture of strictly positive Gaussian densities, continuous jointly in the starting and target state. On any two fixed compact intervals it has a positive lower bound. Every compact set is therefore small, the chain is Lebesgue irreducible and aperiodic, and the geometric Foster–Harris theorem applies (Hairer & Mattingly, 2011). To localize the drift, pick $q(p) < q' < 1$ and absorb its additive constant inside a sufficiently large level set of $1 + |x|^p$.

For the positive case write $a_j = |A_j|$, $P_n = \prod_{j=1}^n a_j$, and $d = \mathbb{E} \log a_1 > 0$. The reverse triangle inequality, iterated, gives

$$\frac{|X_n|}{P_n} \geq |X_0| - \sum_{j=1}^n \frac{R + |N_j|}{P_j}.$$

Fix $0 < c < d$. The strong law implies $\log P_n - cn \rightarrow \infty$, so $\inf_{n \geq 0} P_n e^{-cn} > 0$ almost surely. There is a deterministic $K > 0$ such that $E_K = \{P_n \geq K e^{cn} \text{ for all } n \geq 0\}$ has positive probability. Conditionally on the entire multiplier sequence, $\mathbb{E}(|N_j| \mid (A_k)_{k \geq 1}) \leq C = \sqrt{2v_+/\pi}$. On E_K , therefore,

$$\mathbb{E} \left[\sum_{j \geq 1} \frac{R + |N_j|}{P_j} \mid (A_k)_{k \geq 1} \right] \leq D := \frac{R + C}{K(e^c - 1)}.$$

For $|X_0| \geq 4D$, conditional Markov inequality gives probability at least $1/2$ that the weighted noise sum is at most $|X_0|/2$. Consequently the probability of $|X_n| \rightarrow \infty$ from every such large start is at least $\mathbb{P}(E_K)/2 > 0$. The everywhere positive one-step density reaches this region from every start with positive probability. An invariant law π would thus give positive stationary probability δ to escape. For any $M < \infty$, Fatou's lemma and stationarity imply

$$\delta \leq \mathbb{E}_\pi \liminf_n \mathbf{1}_{\{|X_n| > M\}} \leq \liminf_n \mathbb{P}_\pi(|X_n| > M) = \pi(|x| > M).$$

Letting $M \rightarrow \infty$ contradicts that π is a probability. $\square$

For the affine case $r = 0$, negative logarithmic drift also gives the classical perpetuity representation $X \stackrel{d}{=} \sum_{n \geq 1} (A_1 \cdots A_{n-1}) N_n$. The products decay exponentially almost surely and Gaussian innovations grow subexponentially, so the series converges absolutely. Splitting off the first pair proves the distributional recursion. Under the additional Kesten–Goldie assumptions (a root $\mathbb{E}|A|^\kappa = 1$, $0 < \mathbb{E}|A|^\kappa \log |A| < \infty$, nonarithmetic $\log |A|$, the required κ -moment of the innovation, and no deterministic common fixed point), the stationary tails have power index κ . This last tail theorem is classical (Kesten, 1973; Goldie, 1991); the theorem above proves the existence/nonexistence assertion needed here directly. Gaussian forcing excludes absorbing point masses and common deterministic fixed points.

A.3 PROOF OF THE REFLECTED-LOG CENTRAL LAW

Let $Z_j = -\log M_j$, $S_n = \sum_{j=1}^n Z_j$, $S_0 = 0$, and $W = \log(r_*/R)$. Then

$$W_{t+1} = \max\{0, W_t + Z_{t+1}\}, \quad \mathbb{E}Z_1 = -\gamma < 0.$$

The strong law implies $S_n \rightarrow -\infty$, hence $W_* := \sup_{n \geq 0} S_n < \infty$ almost surely. Splitting the supremum at its first increment yields $W_* \stackrel{d}{=} \max\{0, Z_1 + W'_*\}$, with an independent copy W'_* . This proves invariance. To see uniqueness and attraction, couple two finite starting values with the same increments. The larger process reaches zero almost surely: before its first visit its value is its starting value plus S_n , which cannot stay positive forever. The smaller process is zero at that time by monotonicity, and the two processes then agree. Couple an arbitrary finite start to an independent stationary start this way. The coupling time is finite almost surely, so its tail tends to zero and total variation convergence follows.

Put $\phi(q) = \mathbb{E}e^{qZ} = m(-q)$. At the positive root $\phi(\alpha) = 1$, define the tilted increment law $F_\alpha(dz) = e^{\alpha z} F(dz)$. Its positive mean is $\mu_\alpha = \mathbb{E}[Ze^{\alpha Z}]$. For $T_u = \inf\{n \geq 1 : S_n > u\}$, the likelihood identity, first at each event $\{T_u = n\}$ and then summed over n , gives

$$\begin{aligned} \mathbb{P}(W_* > u) &= \sum_{n \geq 1} \mathbb{E}_\alpha[e^{-\alpha S_n} \mathbf{1}_{\{T_u = n\}}] \\ &= e^{-\alpha u} \mathbb{E}_\alpha e^{-\alpha O_u}, \quad O_u = S_{T_u} - u. \end{aligned}$$

Under the tilted law $T_u < \infty$ almost surely, since its drift is positive. The nonarithmetic overshoot renewal theorem, with the stated exponential-neighborhood and finite-mean assumptions, gives $O_u \Rightarrow O_\infty$, where O_∞ is finite and nonnegative. The bounded continuous function $e^{-\alpha o}$ yields $\mathbb{E}_\alpha e^{-\alpha O_u} \rightarrow C := \mathbb{E}_\alpha e^{-\alpha O_\infty} \in (0, 1]$. This is the classical Cramér–Lundberg asymptotic; its random-walk renewal formulation is also described by Blanchet & Glynn (2006). Substitution of $u = \log(r_*/r)$ proves the small-ball law. Strict versus weak inequalities have the same asymptotic by squeezing between radii r and $(1 + \epsilon)r$ and then letting $\epsilon \downarrow 0$.

Here is a proof of the concrete density condition in the theorem. Suppose $Z = -\log M$ has a bounded continuous density f , supported in $[-b_-, b_+]$, with $b_+, b_- > 0$. Let τ_+ be its first strictly

positive ladder epoch and define the defective ladder-height measure $G(D) = \mathbb{P}(\tau_+ < \infty, S_{\tau_+} \in D)$ on $(0, b_+]$. Write $p = G((0, b_+])$. Since $S_n \rightarrow -\infty$, the walk has finitely many strict upper records almost surely. The strong Markov property would produce infinitely many records if $p = 1$, so $p < 1$. Decomposing the supremum by its number of records gives

$$\mathcal{L}(W_*) = (1 - p) \sum_{n \geq 0} G^{*n}.$$

The ladder measure has density

$$g_G(x) = \mathbf{1}_{\{0 < x \leq b_+\}} \sum_{n \geq 0} \mathbb{E}[\mathbf{1}_{\{\tau_+ > n\}} f(x - S_n)].$$

The summand is zero unless $-b_+ \leq S_n \leq 0$. A bounded-increment random walk with negative mean has finite expected occupation of this compact interval: for large n , the event $S_n \geq -b_+$ is an exponentially unlikely deviation from its negative mean, by Hoeffding's inequality. Thus g_G is bounded. Dominated convergence using the same finite occupation measure makes it continuous on $(0, b_+)$, with finite one-sided endpoint limits. It is consequently directly Riemann integrable and compactly supported.

Exponential tilting at the stopped ladder epoch gives $\hat{G}(dx) = e^{\alpha x} G(dx)$, which is a probability: under the tilted positive-drift walk the first positive ladder epoch is finite almost surely. Its mean $\hat{\mu} = \int x \hat{G}(dx)$ lies in $(0, b_+]$, and it has a bounded, directly Riemann integrable density $\hat{g}$. Let $\hat{U} = \sum_{n \geq 0} \hat{G}^{*n}$. The positive-increment key renewal theorem applied to $\hat{g}$ gives

$$\hat{u}(w) := \int \hat{g}(w - y) \hat{U}(dy) \rightarrow \frac{1}{\hat{\mu}}.$$

This convolution is a density of $\hat{U} - \delta_0$, since $\hat{U} = \delta_0 + \hat{G} * \hat{U}$. Undoing the tilt in the supremum decomposition therefore gives a density on $(0, \infty)$,

$$g_{W_*}(w) = (1 - p)e^{-\alpha w} \hat{u}(w) \sim \frac{1 - p}{\hat{\mu}} e^{-\alpha w}.$$

Its integral tail identifies the earlier constant as $C = (1 - p)/(\alpha \hat{\mu})$. Finally $p_R(r) = g_{W_*}(\log(r_*/r))/r$, proving the density assertion. The atom $1 - p$ at $W_* = 0$, equivalently at $R = r_*$, is retained. This argument derives the local density from the positive-increment renewal theorem and does not differentiate a CDF asymptotic.

For the harmonic comparison, let $\psi(q) = \log \phi(q)$. It is strictly convex for a nonconstant gain, with $\psi(0) = 0$ and $\psi'(0) = -\gamma < 0$. Its unique positive zero is α . It is negative on $(0, \alpha)$ and positive beyond α within its finite domain. Since $\psi(1) = -\log H$, this proves (8). Existence of the root is a substantive assumption: if $M > 1$ almost surely then no positive inverse root exists.

A.4 A SOLVABLE RETURN MODEL WITH AN EXACT DENSITY

Proposition 7 (Exponential log excursions). *Let $U \sim \text{Exp}(b)$, $V \sim \text{Exp}(c)$ be independent, $b > c > 0$, and $M = e^{V-U}$. In (6), set $\alpha = b - c$ and $\rho = c/b$. Then*

$$\pi_R = (1 - \rho)\delta_{r_*} + \rho \frac{\alpha}{r_*} \left(\frac{r}{r_*} \right)^{\alpha-1} \mathbf{1}_{(0, r_*)}(r) dr. \quad (21)$$

The local logarithmic drift is $1/c - 1/b > 0$ and $m(-q) = bc/[(b - q)(c + q)]$ for $0 \leq q < b$. Thus $m(-\alpha) = 1$, and, when $b > 1$, $H > 1$ precisely when $\alpha > 1$.

Proof. For the reflected coordinate, propose $W \sim (1 - \rho)\delta_0 + \rho \text{Exp}(\alpha)$, independent of fresh U, V . Its Laplace transform is

$$\mathbb{E}e^{-sW} = 1 - \rho + \rho \frac{\alpha}{\alpha + s} = \frac{\alpha(b + s)}{b(\alpha + s)}.$$

Multiplying by the Laplace transform $b/(b + s)$ of U shows $W + U \sim \text{Exp}(\alpha)$. Therefore, for $w \geq 0$,

$$\mathbb{P}((W + U - V)_+ > w) = e^{-\alpha w} \mathbb{E}e^{-\alpha V} = \rho e^{-\alpha w}.$$

This is the proposed invariant law; uniqueness follows from the preceding coupling argument. Transforming back gives (21). Independence gives the displayed moment formula, whose nonzero root is $b - c$. Finally $H = (b - 1)(c + 1)/(bc)$, so $H - 1 = (\alpha - 1)/(bc)$. $\square$

The boundary atom is real and must be retained in plots or simulations. A fair independent sign gives a symmetric invariant coordinate with half the mass on each side, but its two boundary atoms are not a theorem about two smooth interior modes in neural-network coordinates.

A.5 FINITE-MEAN REINJECTION: A USEFUL GENERALIZATION

The hard cap can be replaced by independent excursions. The following version uses bounded increments and entries to make the renewal averaging step transparent. Suppose $Y = \log M$ has a bounded continuous density with bounded support, $\mathbb{E}Y > 0$, $\mathbb{P}(Y < 0) > 0$, and $\mathbb{E}e^{-\alpha Y} = 1$ for $\alpha > 0$. Let the independent entry Z lie in $[-L, -\ell]$, with $0 < \ell \leq L < \infty$. Run $Q_n = Z + \sum_{j=1}^n Y_j$ until $\tau = \inf\{n \geq 0 : Q_n \geq 0\}$; discard the exit and restart at a fresh independent Z . Then

$$\pi_Q(D) = \frac{\mathbb{E} \sum_{n=0}^{\tau-1} \mathbf{1}_{\{Q_n \in D\}}}{\mathbb{E}\tau} \quad (22)$$

is invariant. To see $\mathbb{E}\tau < \infty$, write $\mu = \mathbb{E}Y > 0$. Since $\{\tau > n\}$ implies $\sum_{j=1}^n Y_j < L$, bounded-increment Hoeffding bounds make the tail summable for $n > 2L/\mu$. For bounded test functions, the expected one-step change summed over a cycle telescopes to the next fresh entry minus the present entry; its expectation is zero. This proves (22) is invariant.

Let $m_\alpha = -\mathbb{E}[Y e^{-\alpha Y}] > 0$ and $D_\alpha = \mathbb{E}[e^{-\alpha Z} - e^{-\alpha Q_\tau}] > 0$. For every fixed $h > 0$, the renewal calculation gives

$$\pi_Q((x, x + h]) \sim \frac{D_\alpha}{\alpha m_\alpha \mathbb{E}\tau} (e^{\alpha h} - 1) e^{\alpha x}, \quad x \rightarrow -\infty. \quad (23)$$

Here is the calculation. For the increment law F , write $U = \sum_{n \geq 0} F^{*n}$. Tilting by $e^{-\alpha y}$ yields a probability F_α with negative drift $-m_\alpha$ and $U(dt) = e^{\alpha t} U_\alpha(dt)$. The nonarithmetic random-walk renewal theorem, applied on fixed intervals and then to continuous test functions on those intervals, gives

$$U_z((x, x + h]) \sim e^{-\alpha z} \frac{e^{\alpha h} - 1}{\alpha m_\alpha} e^{\alpha x}.$$

This convergence is uniform over z in a compact set. One can obtain uniformity from convergence on a finite grid of interval endpoints and monotone upper/lower interval approximations. The original entry is in a compact interval, and the exit overshoot lies in $[0, y_+]$, where y_+ bounds the positive increments. Thus the averaging is valid without any unbounded-entry interchange. The strong Markov property gives the exact killed-potential identity

$$U_Z(D) = \mathbb{E} \sum_{n < \tau} \mathbf{1}_{\{Q_n \in D\}} + \mathbb{E} U_{Q_\tau}(D).$$

Subtract the two uniformly valid asymptotics and divide by $\mathbb{E}\tau$ to obtain (23). Bounded local renewal masses give a uniform exponential bound for the fixed-width intervals; a geometric-series domination then permits summing toward $-\infty$. Hence for $R = e^Q$,

$$\mathbb{P}_\pi(R \leq r) \sim \frac{D_\alpha}{\alpha m_\alpha \mathbb{E}\tau} r^\alpha.$$

The bounded continuous increment density also verifies the density version here directly. The tilted unrestricted potential satisfies $U_\alpha = \delta_0 + F_\alpha * U_\alpha$. Its absolutely continuous part therefore has the continuous density $u_\alpha(x) = \int f_\alpha(x - y) U_\alpha(dy)$. The fixed-interval renewal theorem gives vague convergence of $U_\alpha(x + dt)$ to dt/m_α as $x \rightarrow -\infty$. Because f_α is continuous and compactly supported, this convolution converges to $1/m_\alpha$. The convergence is uniform under bounded shifts. At sufficiently negative locations the compactly supported entry and exit measures have no atoms; the killed-potential subtraction therefore has a density there. Undoing the tilt and subtracting its entry and exit coefficients proves $p_R(r) \sim D_\alpha r^{\alpha-1}/(m_\alpha \mathbb{E}\tau)$. This result proves an excursion law with specified return. Applying it to a nonlinear SGD path requires verifying its return and approximation hypotheses, including the distribution of entry near zero.

A.6 PROOF OF THE SMOOTH NONLINEAR SGD EXAMPLE

For the general parameters in Theorem 4, expansion of the sampled loss gives

$$\ell(x; h, e) = \frac{1}{2}(h - \beta)x^2 + \frac{\beta}{2}\log(1 + x^2) - \sqrt{h}ex + \frac{1}{2}e^2.$$

Because $h - \beta \geq \mu - s > 0$, every realized loss is coercive. Taking expectations gives $L(x) = \mu x^2/2 + (\beta/2)\log(1 + x^2) + C$. Its derivative is $x[\mu + \beta/(1 + x^2)]$, so zero is the unique minimizer. Its second derivative is

$$L''(x) = \mu + \beta \frac{1 - x^2}{(1 + x^2)^2}.$$

The fraction attains its minimum $-1/8$ at $x^2 = 3$; thus $\min L'' = \mu - \beta/8 < 0$. Differentiation of the sampled loss gives the exact step

$$X^+ = (1 - \eta(h - \beta))X - \frac{\eta\beta X}{1 + X^2} + \eta\sqrt{h}e.$$

The remainder has absolute value at most $\eta\beta/2$. The assumptions on η imply $a_* := \max\{|1 - \eta(\mu - s)|, |1 - \eta(\mu + s)|\} < 1$. For any $p > 0$, choose $\delta > 0$ so $q = (1 + \delta)a_*^p < 1$. The elementary bound

$$|u + v|^p \leq (1 + \delta)|u|^p + C_{p,\delta}|v|^p$$

(or subadditivity when $p \leq 1$) gives $P|x|^p \leq q|x|^p + C_p$, since the remainder is bounded and the Gaussian noise has all moments. At $p = 1$ this drift, combined with the Gaussian compact minorization in the proof of Proposition 6, gives recurrence, uniqueness, and convergence. This direct argument also covers a zero outer slope, which was excluded from the logarithmic-sign proposition. Iteration from any fixed finite start bounds $\sup_n \mathbb{E}|X_n|^p$. Convergence in law and the Portmanteau theorem give $\int |x|^p \pi(dx) < \infty$, without assuming this moment when integrating the drift.

The conditional noise variances are $\eta^2\sigma^2h$, bounded above and away from zero. Invariance writes the density as a mixture of translated Gaussian densities. Every Gaussian derivative of fixed order is uniformly bounded over its translation and this compact variance interval. Differentiating under the invariant mixture shows that the density is bounded and smooth; positivity is immediate. Reflection equivariance and uniqueness give evenness. Finally the exact sampled Jacobian at zero is $1 - \eta h$. At the numerical instance in the theorem, the signed gains are $-.91425, -1.11575$, whose product is greater than one. Their harmonic mean is greater than one by direct rational arithmetic. Strict convexity of $q \mapsto \mathbb{E}|1 - \eta h|^{-q}$, its negative derivative at zero, and its divergence as $q \rightarrow \infty$ (from the contracting atom) give a unique positive root. Evaluating this finite two-term function gives (13); the signs are exact and do not depend on the displayed decimal rounding.

Increasing β while holding (μ, s, η) fixed leaves the outer slopes unchanged and makes both central gains grow without bound. Every fixed finite β still has a bounded remainder and finite Gaussian moments. Consequently recurrence and all finite polynomial moments can coexist with arbitrarily large central logarithmic and harmonic gains, although a contracting central atom and hence a finite inverse root eventually disappear.

A contrasting occupied inverse moment at a larger step. The same fixed loss at $\eta = 8$ has outer slopes $.76, -.04$ and central Jacobians $-6.6, -7.4$, so all recurrence conclusions remain valid and $H_0 = 6.977142857 \dots > 1$.

The Jacobian at an occupied state is instead

$$D_h(x) = 1 - 8(h - .92) - 7.36 \frac{1 - x^2}{(1 + x^2)^2}.$$

It is state dependent. In fact its simple zeros, together with the strictly positive invariant density, imply $\mathbb{E}_{\pi,h}|D_h(X)|^{-1} = \infty$: the integrand is comparable to $1/|x - x_c|$ near any simple zero x_c . Thus the occupied frozen harmonic gain can be zero while the central logarithmic gain exceeds one. A finite Monte Carlo inverse moment need not detect this divergence. This reinforces why all scalar directions and locations must be identified in an empirical threshold claim.

When central depletion gives off-center modes. Suppose a signed invariant law has a continuous even density f on $\mathbb{R}$, $f(0) = 0$, $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$, and positive total mass. Some $x_0 > 0$ has $f(x_0) > 0$. The limit at zero and infinity confines the supremum on $[0, \infty)$ to a compact subinterval of $(0, \infty)$, where continuity attains it. Its negative reflection is another maximizer. Thus there are at least two off-center modes. This does not exclude additional modes or a plateau. Exactly two requires, for example, a unique maximum on each half-line. Moreover, at fixed nonzero additive Gaussian noise the density in the theorem is positive at zero, so an ideal multiplicative depletion law cannot be transferred literally down to zero without a small-noise argument.

A.7 A COUNTEREXAMPLE TO A UNIVERSAL HARMONIC-TO-BIMODALITY CLAIM

On $(0, 1)$, take $M = 1/2$ with probability $1/5$ and $M = 2$ otherwise. If $MR < 1$, set $R' = MR$; otherwise independently restart from density

$$g(r) = \begin{cases} (24 + 21r^2)/41, & 0 < r < 1/2, \\ (72 - 27r^2)/41, & 1/2 \leq r < 1. \end{cases}$$

The multiplier has $\gamma = (3/5) \log 2 > 0$, $H = 5/4 > 1$, and inverse root $\alpha = 2$. Nevertheless the exact invariant density is

$$f(r) = \frac{12}{11}(1 - r^2/4), \quad 0 < r < 1,$$

which decreases from the origin. Both displayed functions integrate to one. The killed transfer operator is

$$Tf(r) = \frac{2}{5}f(2r)\mathbf{1}_{\{r < 1/2\}} + \frac{2}{5}f(r/2).$$

Direct substitution on the two half-intervals gives $f - Tf = (41/110)g$. Under f , the reset probability is $(4/5) \int_{1/2}^1 f(r)dr = 41/110$, so $f = Tf + (41/110)g$ proves invariance. Adding a reflection-symmetric sign yields a smooth central maximum. The return law places substantial mass arbitrarily near zero and does not satisfy the inverse entry-moment condition of the more general regenerative theorem. The multiplier alone cannot distinguish this counterexample from a regular-return model with the same α .

A.8 EXACTLY TWO MODES IN THE CONFINED DIFFUSION NORMAL FORM

For clarity, one setting where mode count can be proved completely is the *diffusion* normal form

$$dY = (aY - cY^3)dt + \sqrt{\sigma^2 + \nu^2 Y^2} dW, \quad c, \sigma, \nu > 0.$$

The quartic inward drift and nondegenerate diffusion give a nonexplosive positive recurrent scalar process. Solving its zero-current equation $(Dp)' = 2(ay - cy^3)p$, where $D = \sigma^2 + \nu^2 y^2$, gives

$$p(y) = C(\sigma^2 + \nu^2 y^2)^{a/\nu^2 + c\sigma^2/\nu^4 - 1} e^{-cy^2/\nu^2}.$$

It is normalizable, and direct differentiation yields

$$(\log p)'(y) = \frac{2y[(a - \nu^2) - cy^2]}{\sigma^2 + \nu^2 y^2}.$$

There is one mode at zero for $a \leq \nu^2$, and exactly two modes at $\pm \sqrt{(a - \nu^2)/c}$ for $a > \nu^2$. The zero-additive-noise local logarithmic exponent is $a - \nu^2/2$. When $a > \nu^2/2$, its nontrivial zero-noise invariant density on the punctured line has central exponent $\alpha = 2a/\nu^2 - 1$, so the mode transition occurs at $\alpha = 1$. This is a diffusion theorem; fixed-step SGD requires its own confinement and approximation argument. In particular, a raw cubic explicit update with nondegenerate Gaussian innovations can escape and does not inherit this invariant law merely because its local Taylor expansion matches the diffusion drift.

B MATRIX RENEWAL AT A LOCALLY EXPANDING CENTER

B.1 DEFINITIONS AND EXPLICIT ASSUMPTIONS

Let $A_1, A_2, \dots$ be iid invertible matrices. Work on a compact communicating class $\mathcal{S}$ of signed unit directions; projective space is sufficient when no statement depends on sign. For a nonzero vector, write $U = r_0 e^Z S$. Define the Markov additive process

$$S_{n+1} = \frac{A_{n+1} S_n}{\|A_{n+1} S_n\|}, \quad Z_{n+1} = Z_n + Y_{n+1}, \quad Y_{n+1} = \log \|A_{n+1} S_n\|. \quad (24)$$

Its joint direction-increment kernel is $K(s, ds', dy)$. The term ‘‘Markov additive’’ means precisely that this kernel depends on the current direction but not on the current log radius. Let P_q be (10), acting on continuous functions with the supremum norm, and let $k(q)$ be its spectral radius. In particular $P_0 1 = 1$ and $k(0) = 1$.

Assumption 1 (Negative-order renewal hypotheses). The following conditions hold.

1. For some $\alpha > 0$ there exist a continuous h , constants $0 < h_{\min} \leq h \leq h_{\max} < \infty$, and a probability measure χ such that

$$P_{-\alpha} h = h, \quad \chi P_{-\alpha} = \chi, \quad \chi(h) = 1, \quad k(-\alpha) = 1. \quad (25)$$

For some $t \in (0, \alpha)$, $P_{-\alpha+t}$ is bounded on $C(\mathcal{S})$ and $k(-\alpha + t) < 1$.

2. The probability kernel

$$\hat{K}(s, ds', dy) = e^{-\alpha y} \frac{h(s')}{h(s)} K(s, ds', dy) \quad (26)$$

has a positive Harris recurrent angular chain, with invariant law $\hat{\chi}(ds) = h(s)\chi(ds)$, and each one-step angular law $\hat{P}(s, \cdot)$ is absolutely continuous with respect to $\hat{\chi}$. Under $\hat{\chi}$,

$$\hat{\mathbb{E}}|Y_1| < \infty, \quad \hat{\mathbb{E}}Y_1 = -m < 0. \quad (27)$$

3. The tilted Markov additive process is nonarithmetic: there is no $b > 0$ and measurable $c : \mathcal{S} \rightarrow [0, b)$ such that $Y_1 \in c(S_0) - c(S_1) + b\mathbb{Z}$ almost surely under its stationary angular law.

These are hypotheses on the actual random matrices, not conclusions from a positive one-step scalar log gain. Absolute continuity in the second condition is a convenient sufficient condition extending the almost-every-start Markov renewal theorem to every starting direction. Singular or finitely supported matrix laws may require a different all-start renewal theorem; the present statement does not silently cover them. If the dominant eigenvalue and eigenfunction are differentiable at $-\alpha$, differentiating $P_q h_q = k(q)h_q$ and integrating against χ gives $k'(-\alpha) = \chi(P'_{-\alpha} h) = -m$. Thus $m = -(\log k)'(-\alpha)$ in that case. Strict convexity of $\log k$ between the two roots guarantees $k(-\alpha + t) < 1$.

Let $\tau = \inf\{n \geq 0 : Z_n \geq 0\}$ and let ζ be an entry law on $(-\infty, 0) \times \mathcal{S}$. We impose

$$L := \mathbb{E}\zeta\tau \in (0, \infty), \quad \mathbb{E}\zeta e^{-\alpha Z_0} < \infty. \quad (28)$$

At the transition that would exit, the regenerative reference chain instead draws its next state from an independent copy of ζ . Consequently each cycle contains states numbered $0, \dots, \tau - 1$. This construction specifies a reference process. Probabilistic regeneration of actual SGD is addressed separately below.

B.2 A UNIFORM OCCUPATION BOUND DERIVED FROM THE SPECTRUM

Lemma 8 (Tilted interval potentials are uniformly bounded). *Under Assumption 1(1), for each $b > 0$ there is $C_b < \infty$ such that*

$$\sup_{z, s, x} \hat{\mathbb{E}}_{z, s} \sum_{n \geq 0} \mathbf{1}_{\{Z_n \in (x, x+b]\}} \leq C_b. \quad (29)$$

Proof. The exact change of measure gives

$$\widehat{\mathbb{E}}_s e^{t(Z_n - Z_0)} = \frac{P_{-\alpha+t}^n h(s)}{h(s)} \leq C \rho^n \quad (30)$$

for some $\rho < 1$ and $C < \infty$. Indeed, choose ρ strictly between $k(-\alpha + t)$ and 1; the spectral-radius formula bounds the operator powers by a constant times ρ^n , and h is bounded above and away from zero. If $Z_0 \in (x, x + b]$, exponential Markov inequality yields

$$\widehat{\mathbb{P}}_{Z_0, s}(Z_n \in (x, x + b]) \leq e^{tb} C \rho^n.$$

Summing over n bounds occupation by $e^{tb} C / (1 - \rho)$. For any other initial state, apply this bound at its first visit to the interval, using the strong Markov property. If there is no first visit the occupation is zero. This proves the claim, uniformly over both direction and log radius. $\square$

B.3 UNRESTRICTED AND KILLED RENEWAL ASYMPTOTICS

Write $U_{z, s}$ and $\widehat{U}_{z, s}$ for the unrestricted occupation measures of (24) under its original and tilted laws. For n steps, multiplying (26) along a path gives the exact likelihood ratio

$$\left. \frac{d\widehat{\mathbb{P}}_{z, s}}{d\mathbb{P}_{z, s}} \right|_{\mathcal{F}_n} = e^{-\alpha(Z_n - z)} \frac{h(S_n)}{h(s)}. \quad (31)$$

Consequently

$$U_{z, s}(dx, ds') = e^{\alpha(x - z)} \frac{h(s)}{h(s')} \widehat{U}_{z, s}(dx, ds'). \quad (32)$$

Apply the Markov renewal theorem of [Alsmeyer \(1997, Theorem 1\)](#) to the positive-drift process $-Z$. For an angular set D and a finite log interval, its conclusion, together with the exponential test weight in (32), gives

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} U_{z, s}((x, x + b] \times D) = e^{-\alpha z} h(s) \frac{e^{\alpha b} - 1}{\alpha m} \chi(D). \quad (33)$$

Initially the theorem gives the limit for $\widehat{\chi}$ -almost every starting direction. To obtain it for an arbitrary s , split the tilted occupation into its time-zero atom and its potential after the first step. The time-zero atom eventually lies outside the translated interval. The next direction has a law absolutely continuous with respect to $\widehat{\chi}$, so the almost-everywhere limit applies after this step. Lemma 8, together with the bounded test weight $e^{\alpha u}/h(s')$, permits dominated convergence over this first step. This proves the limit for every s and every finite initial z . The angular factor is χ , since division by $h(s')$ cancels the $h(s')$ in $\widehat{\chi}$.

Moreover, (32) and the same lemma give a constant C'_b such that, for all x, z, s ,

$$e^{-\alpha x} U_{z, s}((x, x + b] \times \mathcal{S}) \leq C'_b e^{-\alpha z} h(s). \quad (34)$$

This bound is the domination needed for integration over entry and exit states. It has been proved, rather than included as an unverified “uniform renewal” assumption.

Proof of Theorem 3. Let G_ζ denote expected occupation before τ . The strong Markov property gives, on bounded log intervals,

$$G_\zeta = U_\zeta - \mathbb{E}_\zeta U_{Z_\tau, S_\tau}. \quad (35)$$

Finite mean exit time implies $\tau < \infty$ almost surely under $\mathbb{P}_\zeta$. The entry moment in (28), bounded h , and $Z_\tau \geq 0$ permit integrating (33) in both terms by (34). Hence

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} G_\zeta((x, x + b] \times D) = D_\alpha \frac{e^{\alpha b} - 1}{\alpha m} \chi(D), \quad (36)$$

where

$$D_\alpha = \mathbb{E}_\zeta [e^{-\alpha Z_0} h(S_0) - e^{-\alpha Z_\tau} h(S_\tau)]. \quad (37)$$

Its sign cannot be read by comparing endpoint radii alone, because directions change. For each fixed start, summing the stopped likelihood ratio (31) over events $\{\tau = n\}$ instead gives

$$\mathbb{E}_{z, s}[e^{-\alpha Z_\tau} h(S_\tau); \tau < \infty] = e^{-\alpha z} h(s) \widehat{\mathbb{P}}_{z, s}(\tau < \infty).$$

Therefore

$$D_\alpha = \mathbb{E}_\zeta \left[e^{-\alpha Z_0} h(S_0) \hat{\mathbb{P}}_{Z_0, S_0}(\tau = \infty) \right]. \quad (38)$$

The stated positive-survival condition makes $D_\alpha > 0$. It is enough for a positive entry mass to be sufficiently far below the upper boundary: by (30),

$$\hat{\mathbb{P}}_{z,s}(\tau < \infty) \leq \sum_{n \geq 1} \hat{\mathbb{P}}_{z,s}(Z_n \geq 0) \leq \frac{C\rho}{1-\rho} e^{tz},$$

which is strictly below one for all sufficiently negative z .

The regenerative occupation formula

$$\pi(F) = \frac{1}{L} \mathbb{E}_\zeta \sum_{n=0}^{\tau-1} \mathbf{1}_{\{(Z_n, S_n) \in F\}} \quad (39)$$

defines an invariant probability. For a bounded test function, summing its expected one-step change over a cycle telescopes to the difference between two cycle starts, both with law ζ ; hence its integral against $\pi P - \pi$ is zero. To pass from intervals to the half-line $(-\infty, x]$, partition it into intervals $(x - (j+1)b, x - jb]$. The bound (34) dominates their rescaled occupations by a summable geometric sequence in j . Summing (36) therefore yields

$$\lim_{x \rightarrow -\infty} e^{-\alpha x} \pi(Z \leq x, S \in D) = \frac{D_\alpha}{\alpha m L} \chi(D). \quad (40)$$

Substituting $x = \log(r/r_0)$ proves the theorem, with $C = D_\alpha/(\alpha m L)$. $\square$

The theorem proves a small-ball asymptotic. Differentiating an asymptotic cumulative distribution without further hypotheses is invalid. A radial-density conclusion requires the corresponding renewal-density result, for example a suitable smoothing condition giving convergence of local densities. Angular density conclusions additionally require regularity of χ .

B.4 A NONCOMMUTING LAW THAT VERIFIES EVERY RENEWAL HYPOTHESIS

The preceding assumptions are realizable even by symmetric random matrices. For $d \geq 2$, take

$$A = M(I - 2vv^\top), \quad v \sim \text{Unif}(S^{d-1}), \quad M \sim \frac{1}{4} \text{Unif} [.49, .51] + \frac{3}{4} \text{Unif} [1.99, 2.01], \quad (41)$$

with independent draws at each update. Independent reflections generically do not commute. Their norm gain is M , so $P_q = m(q)R$, where $m(q) = \mathbb{E} M^q$ and R is the angular reflection kernel. Thus $h = 1$, $k(q) = m(q)$, and χ is uniform surface probability. This scalar formula is special to this example; the matrix theorem itself retains the coupled dynamics.

For completeness the reflection kernel has, with respect to uniform surface probability, density

$$R(s, ds') = \|s - s'\|^{2-d} \chi(ds') \quad (s' \neq s). \quad (42)$$

To verify it, put $s = e_1$ and let θ be the angle between s and s' . Reflection gives $|v^\top s| = \sin(\theta/2)$. Changing variables from the spherical coordinate of v gives the angle density $c_d \cos^{d-2}(\theta/2)$, whereas a uniform s' has angle density $c_d \sin^{d-2} \theta$. Their ratio is $(2 \sin(\theta/2))^{2-d} = \|s - s'\|^{2-d}$. In particular $R(s, \cdot) \geq 2^{2-d} \chi(\cdot)$, so it is a uniformly mixing Harris kernel, and its symmetry verifies invariance of χ . Its one-step laws are absolutely continuous.

The top product norm equals $\prod_{j=1}^n M_j$, since every remaining factor is orthogonal. Hence

$$\gamma = \mathbb{E} \log M \geq \frac{1}{4} \log(.49) + \frac{3}{4} \log(1.99) > 0, \quad m(-1) \leq \frac{1}{4(.49)} + \frac{3}{4(1.99)} < 1.$$

Strict log convexity and the positive mass below one imply a unique negative root $m(-\alpha) = 1$ with $\alpha > 1$ and $m(-\alpha + t) < 1$ for every $t \in (0, \alpha)$. Tilting changes only the scalar multiplier law; it leaves R unchanged. The tilted log increment has bounded support, a density, and strictly negative mean $m'(-\alpha) < 0$, so all spectral and nonarithmetic hypotheses hold.

Choose the entry log radius $Z_0 = -1$ and an independent uniform direction. Since the original log increments are iid with mean $\gamma > 0$ and are bounded above by $\log 2.01$, Wald's identity at $\tau \wedge n$ gives

$$\gamma \mathbb{E}(\tau \wedge n) = \mathbb{E}Z_{\tau \wedge n} + 1 \leq \log 2.01 + 1.$$

The strong law gives almost-sure exit; monotone convergence now gives finite mean exit time. The entry inverse moment is e^α . Under the tilted law $e^{\alpha Z_n}$ is a nonnegative martingale; optional stopping yields $\hat{\mathbb{P}}_{-1,s}(\tau < \infty) \leq e^{-\alpha} < 1$. The central-law coefficient is therefore strictly positive. All hypotheses of Theorem 3 have been verified for this noncommuting law.

The matrices are symmetric, so they can be written $A = I - \eta H$ for symmetric sample Hessians. In $d = 2$, $\mathbb{E}(I - 2vv^\top) = 0$, so $\mathbb{E}H = I/\eta$ is positive definite even though individual sample Hessians need not be. The independent restart remains an explicit return mechanism of the reference process; it is not silently identified with an SGD update.

B.5 AN EXACT, CONDITIONAL TRANSFER STATEMENT FOR NONLINEAR SGD

Proposition 9 (Nonlinear occupation transfer). *Consider a nonlinear chain that has genuine regenerative cycles of finite mean length T , with cycle endpoints outside the local domain $\mathcal{D} = \{(z, s) : z < 0\}$. Let Q be its actual transition kernel killed on leaving $\mathcal{D}$, and let L_0 be the kernel of (24) killed at its upper exit. Let G be the expected measure of all local visits in one complete cycle, and ν the expected measure of every entry into $\mathcal{D}$ in that cycle. Put*

$$\mathcal{J} = \nu + G(Q - L_0), \quad H(z, s) = e^{-\alpha z} h(s) \hat{\mathbb{P}}_{z,s}(\tau = \infty). \quad (43)$$

Assume the killed linear chain exits from G -almost every starting point and that

$$\int e^{-\alpha z} |\mathcal{J}|(dz, ds) < \infty, \quad D_Q := \int H d\mathcal{J} > 0. \quad (44)$$

Under Assumption 1, the genuine invariant law has the same central exponent:

$$\lim_{r \downarrow 0} \frac{\pi(\|U\| \leq r, S \in D)}{(r/r_0)^\alpha} = \frac{D_Q}{\alpha m \mathbb{E}T} \chi(D). \quad (45)$$

Proof. Counting each visit as either an entry or the successor of a previous local visit yields $G = \nu + GQ$. Thus $G = \mathcal{J} + GL_0$ exactly, and

$$\sum_{j=0}^{n-1} \mathcal{J} L_0^j = G - GL_0^n.$$

Since G is finite and the killed linear chain exits G -almost everywhere, GL_0^n tends to zero in total variation. For each fixed bounded log interval, the killed potential is bounded by its unrestricted potential, and (34) shows absolute integrability against $|\mathcal{J}|$ by (44). We may therefore pass to the infinite sum and then integrate the killed asymptotic against the signed source. The coefficient is D_Q . Summing intervals as in the preceding proof and dividing occupation by the complete nonlinear cycle length proves (45). $\square$

The source is signed. A nonnegative entry law does not guarantee $D_Q > 0$, because nonlinear correction can cancel its leading term. The source condition can be checked without assuming the desired central exponent. For example, suppose $V_p(z, s)$ is comparable to e^{-pz} for some $0 < p < \alpha$ and satisfies

$$QV_p \leq \rho_p V_p, \quad \rho_p < 1, \quad \nu(e^{-\alpha z}) < \infty, \quad (46)$$

and

$$\int e^{-\alpha z'} |Q - L_0|((z, s), d(z', s')) \leq CV_p(z, s). \quad (47)$$

Decomposing the cycle into its actual local excursions gives $G = \nu \sum_{n \geq 0} Q^n$ and hence $G(V_p) \leq \nu(V_p)/(1 - \rho_p) < \infty$. Equations (46)–(47) then imply (44)'s integrability assertion. An eigenfunction h_p with $P_{-p}h_p = k(-p)h_p$ and $k(-p) < 1$ suggests $V_p = e^{-pz}h_p(s)$ for this independent drift calculation.

A Taylor expansion $F_B(u) = A_B u + O(\|u\|^{1+\delta})$ does not by itself prove (47): nearby point masses have order-one total variation distance. Smooth transition laws or a different perturbative renewal theorem are needed. Fixed nondegenerate additive noise also destroys the multiplicative approximation as $r \downarrow 0$. For that case an intermediate small-noise annulus, with estimates uniform in the noise scale, must replace the literal zero-radius limit. We do not claim these transfer conditions for the neural experiments.

C AN EXACT SGD CONSTRUCTION AND A PROJECTION COUNTEREXAMPLE

This example verifies stationarity and positive local matrix-product growth in every dimension. It also rules out a universal inference from matrix negative moments to fixed-coordinate bimodality.

Proposition 10 (Stationarity past the local matrix thresholds). *For every $d \geq 2$ and $\rho, \sigma > 0$, there is an explicit online nonquadratic regression loss for which fixed-step SGD has a unique attracting invariant probability with finite second moment, while its linearization at the population minimizer satisfies*

$$\gamma \geq \frac{\log 2}{2d} > 0, \quad k(-1) \leq 1 - \frac{1}{8d} < 1, \quad k(-d) = \frac{7}{8} < 1. \quad (48)$$

The pressure has a unique nonzero root $k(-\alpha) = 1$ with $\alpha > d$. Nevertheless, for $d \geq 3$ every fixed one-dimensional projection of the invariant law has a centrally unimodal density.

Proof. At each update independently sample

$$v \sim \text{Unif}(S^{d-1}), \quad h = \begin{cases} 1/2 & \text{with probability } 1/4, \\ 3 & \text{with probability } 3/4, \end{cases} \quad e \sim N(0, \sigma^2).$$

Let $a = \sqrt{h} v$, $q = (a^\top u)^2/2$, and use step size one on

$$\ell(u; a, e) = \frac{1}{2}(a^\top u - e)^2 - \frac{1}{2}\{q - \rho \log(1 + q/\rho)\}. \quad (49)$$

The centered residual makes the population loss, apart from a constant, the expectation of $q/2 + (\rho/2) \log(1 + q/\rho)$. This expression is positive when $q > 0$, and the random directions span the space, so the unique population minimizer is zero. Differentiation gives the exact SGD map

$$u^+ = \left(I - h \left[1 - \frac{q}{2(\rho + q)} \right] v v^\top \right) u + e \sqrt{h} v. \quad (50)$$

Local matrix growth. The derivative at zero is $A = I - h v v^\top$. For any fixed unit s , set $T = (v^\top s)^2$, so $\mathbb{E}T = 1/d$. The norm gain is

$$G^2 = \|As\|^2 = 1 + h(h - 2)T, \quad \frac{1}{2} \leq G \leq 2.$$

Isotropy makes this conditional gain law independent of the entire past, including the evolving direction of a matrix product. Thus its log gains are iid. The chord inequalities

$$\frac{1}{2} \log(1 + 3T) \geq T \log 2, \quad \frac{1}{2} \log(1 - 3T/4) \geq -T \log 2$$

give $\mathbb{E} \log G \geq (\log 2)/(2d) > 0$. For each deterministic nonzero initial vector the strong law identifies the product exponent with this expectation. Applying this simultaneously to a finite orthonormal basis, and bounding the operator norm above by the Frobenius norm and below by each column norm, proves that the top matrix-product exponent has the same value.

Negative pressure. For every real z , isotropy gives $P_z 1 = m(z)1$, where $m(z) = \mathbb{E}G^z$. Positivity yields $\|P_z\|_\infty = m(z)$, so $k(z) = m(z)$ exactly. The convex function

$$F(t) = \frac{1}{4}(1 - 3t/4)^{-1/2} + \frac{3}{4}(1 + 3t)^{-1/2}$$

has $F(0) = 1$ and $F(1) = 7/8$, hence $F(t) \leq 1 - t/8$ for $0 \leq t \leq 1$. Consequently $k(-1) = \mathbb{E}F(T) \leq 1 - 1/(8d)$. For any invertible matrix B , polar integration of $e^{-\|Bx\|^2}$ gives

$$\mathbb{E}_{s \sim \text{Unif}(S^{d-1})} \|Bs\|^{-d} = |\det B|^{-1}. \quad (51)$$

Since $|\det(I - h v v^\top)| = |1 - h|$, isotropy and (51) imply $k(-d) = \frac{1}{4} \cdot 2 + \frac{3}{4} \cdot \frac{1}{2} = 7/8$. All moments exist because $G \in [1/2, 2]$. The function $\log m$ is strictly convex because $\log G$ is nonconstant; $m(0) = 1$ and $m'(0) = \gamma > 0$. Since $G < 1$ with positive probability, $m(-s) \rightarrow \infty$ as $s \rightarrow \infty$. There is therefore a unique nonzero negative root, and $m(-d) < 1$ puts it at $-\alpha < -d$.

Global nonlinear return. Rewrite (50) exactly as

$$u^+ = A_\infty u + b(u; h, v) + e\sqrt{h}v, \quad A_\infty = I - \frac{h}{2}vv^\top, \quad (52)$$

$$b(u; h, v) = -\frac{\rho h}{2(\rho + q)}vv^\top u, \quad \|b(u; h, v)\| \leq B_0 := \sqrt{3\rho/8}. \quad (53)$$

The bound follows by maximizing $x/(\rho + hx^2/2)$ for $x \geq 0$. With $c = h/2$, $\mathbb{E}[c(2 - c)] = 43/64$, and hence

$$\mathbb{E}\|A_\infty u\|^2 = \theta\|u\|^2, \quad \theta = 1 - \frac{43}{64d} < 1.$$

The centered independent residual removes cross terms and contributes $\mathbb{E}\|e\sqrt{h}v\|^2 = 19\sigma^2/8$. Young's inequality gives

$$\mathbb{E}[\|u^+\|^2 \mid u] \leq (1 + \epsilon)\theta\|u\|^2 + (1 + \epsilon^{-1})B_0^2 + 19\sigma^2/8.$$

Choosing $\epsilon = (1 - \theta)/(2\theta)$ makes the first coefficient $\lambda = (1 + \theta)/2 < 1$. This is a geometric drift bound for $V(u) = 1 + \|u\|^2$. The map is Feller. Iterating the bound gives uniform second moments for the time marginals from any fixed initial state; their Cesàro averages are tight, and any weak limit is invariant. Lower semicontinuity gives a finite invariant second moment.

Irreducibility and uniqueness. Conditional on h , the one-step kernel has, for $y \neq u$, density

$$p_h(u, y) = \frac{2\varphi_{\sigma\sqrt{h}}(r - \mu_h(w, u))}{|S^{d-1}|r^{d-1}}, \quad r = \|y - u\|, \quad w = (y - u)/r, \quad (54)$$

where φ_s is the centered Gaussian density of standard deviation s and

$$\mu_h(w, u) = -h \left[1 - \frac{h(w^\top u)^2/2}{2(\rho + h(w^\top u)^2/2)} \right] w^\top u.$$

To derive (54), condition on the random axis and write the signed displacement along it as a Gaussian with mean $\mu_h(w, u)$ and variance $\sigma^2 h$. The two signed axes $w, -w$ give equal contributions, and polar coordinates give the displayed Jacobian. The density is strictly positive. On compact source and target sets, r and the absolute Gaussian argument are uniformly bounded, so it has a positive uniform lower bound off the diagonal; the diagonal is Lebesgue-null. Every compact set is therefore small, the chain is Lebesgue-irreducible, and minorizing a ball on itself gives aperiodicity. The drift and small-set condition imply unique positive Harris recurrence and convergence from every starting point (Hairer & Mattingly, 2011).

Fixed projections remain unimodal. The update law commutes with every orthogonal transformation. Uniqueness therefore makes its invariant distribution rotationally invariant. We may write $U = RS$, where S is uniform on the sphere independently of $R > 0$. For any unit direction v_0 and $d \geq 3$, the conditional density of $v_0^\top U$ is

$$f(x \mid R) = \frac{c_d}{R} \left(1 - \frac{x^2}{R^2} \right)^{(d-3)/2} \mathbf{1}_{\{|x| < R\}}. \quad (55)$$

It is symmetric and nonincreasing in $|x|$: it is uniform on $(-R, R)$ when $d = 3$. Mixing over R preserves this monotonicity. Thus every fixed projection is centrally unimodal despite (48) and $\alpha > d$. $\square$

This proposition establishes post-local-Lyapunov stationarity without claiming the central renewal law for the noisy chain. In particular, fixed additive noise cannot be neglected in the limit $\|u\| \downarrow 0$. The sample loss in this particular example is nonquadratic but convex along its observed coordinate: writing $x = a^\top u$ and $t = x^2/(2\rho)$, its second derivative is $1/2 + (1 - t)/(2(1 + t)^2) \geq 7/16$. It should not be used as the separate nonconvex example.

D PROOF OF THEOREM 5: AN EXACT POSITIVE-EXPONENT CONTROL

A randomly perturbed tent map can be realized as SGD on coercive sampled losses. This control gives an analytic coexistence result for globally C^1 , piecewise- C^2 losses; it is not a novelty claim about random expanding maps or a proof of the smooth neural examples' measured signs. Let $a_t \in \{0.58, 0.62\}$ be iid equiprobable and let $\xi_t \sim \text{Unif}[-0.38, 0.38]$ be iid and independent of the a_t . Define

$$\Psi(x) = \begin{cases} x - x|x|, & |x| \leq 1, \\ -x + \text{sign}(x), & |x| > 1, \end{cases} \quad \ell(x; a, \xi) = \frac{x^2}{2} - a\Psi(x) - \xi x.$$

Values and first derivatives match at $-1, 0, 1$, so these losses are C^1 with globally Lipschitz gradients, but not globally C^2 . Every sample loss is coercive. At step size one its exact SGD update is

$$X_{t+1} = F_{a_t}(X_t) + \xi_t, \quad F_a(x) = a\{1 - 2\min(|x|, 1)\}.$$

No projection, gradient clipping, or reset is applied to the trajectory; the bounded map is the gradient update of the displayed piecewise loss. The population loss $L = x^2/2 - 0.6\Psi(x)$ has unique minimizer $3/11$ and curvature -0.2 on $(-1, 0)$, hence is nonconvex.

Invariant law and an explicit mixing bound. Every state enters $K = [-1, 1]$ in one update. Write P for the transition kernel, $I = [-.5, .5]$, and $J = [.27, .35]$. A first-step uniform-noise interval has center in $[-.62, .62]$ and intersects I in length at least $.26$. For $y \in I$, both possible $F_a(y)$ lie in $[0, .62]$; every uniform-noise interval with such a center contains J . Thus, globally in x ,

$$P^2(x, dz) \geq \frac{.26}{.76^2} \mathbf{1}_J(z) dz = \epsilon \frac{\mathbf{1}_J(z)}{.08} dz, \quad \epsilon = \frac{.08 \cdot .26}{.76^2} = 0.03601108033 \dots$$

The Doeblin contraction for P^2 gives a unique invariant probability π and, for every initial law ν ,

$$\|\nu P^n - \pi\|_{\text{TV}} \leq (1 - \epsilon)^{\lfloor n/2 \rfloor},$$

where total variation is $\sup_A |\nu(A) - \pi(A)|$. Uniqueness for P^2 also gives $\pi P = \pi$. The one-step density is bounded by $1/.76$; therefore so is the invariant density. Compact support gives all moments and finite expected risk. Smoothness of this density is not asserted.

Exact stationary common-noise growth. Absolute continuity implies that a stationary trajectory almost surely never visits $\{-1, 0, 1\}$. At all its differentiability points, $F'_a(x) = 2a$ for $-1 < x < 0$ and $F'_a(x) = -2a$ for $0 < x < 1$. Hence the strong law gives exactly

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left| \prod_{t=0}^{n-1} F'_{a_t}(X_t) \right| = \frac{1}{2} \log(1.16 \cdot 1.24) = 0.18176569237 \dots > 0.$$

Both the curvature randomness and the additive forcing are genuine. At the population minimizer the frozen derivatives are also $-2a$; its harmonic gain is $1.16 \cdot 1.24/1.20 > 1$. Thus central expansion, positive stationary tangent growth, uniform attraction in distribution, and finite risk coexist in this explicit control. No mode-count claim is needed.

An essential initialization qualification. For $|X_0| > 1$ the first derivative is exactly zero. A tangent transported from that original initial state remains zero and has exponent $-\infty$. After one update the physical state is in the interior of K and off all kinks almost surely; a tangent initialized *then* has the positive exponent above. Such a post-entry diagnostic must be distinguished from the original derivative product. A kink initial point has no classical first-step derivative. This distinction changes no physical SGD update. Common-noise derivative checks share both a_t and ξ_t .

Random tent maps are established examples in the theory of random interval systems; see Kalle and Maggioni, *Invariant densities for random systems of the interval*, Ergodic Theory and Dynamical Systems 42 (2022), 141–179, doi:10.1017/etds.2020.127 (arXiv:1805.11430). Their work is background here: the explicit minorization and derivative calculation prove this control directly.

An open step-size region for the same loss. Keep the loss and noise laws fixed and take $\eta \in [.98, 1]$. The map is $G_{\eta,a,\xi}(x) = (1 - \eta)x + \eta F_a(x) + \eta\xi$. With $q = 1 - \eta$, convexity of K gives invariance and the global bound $d_K(G(x)) \leq qd_K(x)$. For $x \in K$ the deterministic center lies in $[-.6276, .62]$, and the noise half-width is at least .3724. Its overlap with I has length at least .2448. For $y \in I$ the next center lies in $[-.01, .62]$, whose noise intervals all contain J . Thus the preceding minorization holds on K uniformly with

$$\epsilon_* = \frac{.08 \cdot .2448}{.76^2} = 0.03390581717 \dots$$

There is a unique invariant probability π_η on K , with all moments and the corresponding uniform TV bound for starts in K . For $\eta < 1$, any finite outside start has distance at most $q^n d_K(X_0)$. Once this is below .1, the independent event $\xi \geq 0$, of probability 1/2, forces entry: both endpoint images lie in $[-.6276, -.176]$, and the outside displacement is at most $.1q \leq .002$. Hence entrance occurs almost surely in finite time. Every invariant law on the full line must be supported on K by the distance contraction, so uniqueness is global. Convergence from every initial law follows, but is not uniform over the unbounded line when $\eta < 1$.

On the two halves of K the derivative magnitudes are respectively $1 + \eta(2a - 1)$ and $\eta(1 + 2a) - 1$. Bounded logarithmic gains and uniform ergodicity give the stationary exponent, with the uniform bound

$$\lambda_\eta \geq \log \min\{1 + .16\eta, 2.16\eta - 1\} \geq \log(1.1168) = 0.11046745303 \dots > 0.$$

The same population minimizer also has frozen gains of magnitude $\eta(1 + 2a) - 1 > 1$. This proves coexistence over an open interval of steps, not just a tuned single step. For $\eta < 1$, the outside derivative is $q > 0$, so finitely many pre-entry factors do not alter the exponent of a tangent started away from a kink. The outside-start annihilation at $\eta = 1$ remains a distinct endpoint qualification.

E WHY THE CONSTRUCTED NEURAL CONTROLS HAVE INVARIANT LAWS

The following argument concerns the explicitly forced, bounded-parameter models in Appendix F.2; it does not assert recurrence for ordinary-network SGD.

Proposition 11 (Bounded data gradients and latent return). *On a finite-dimensional active parameter space, consider the exact update*

$$\begin{aligned} \theta^+ &= a\theta + b_B(\theta) + \nu Z, & a &= 1 - \eta\mu, & |a| < 1, \\ \sup_{B,\theta} \|b_B(\theta)\| &\leq K < \infty, & Z &\sim N(0, I), & \nu > 0. \end{aligned}$$

Assume fresh iid batches and Gaussian vectors, independent of each other and of the current state, and continuous b_B . The chain has a unique attracting invariant probability, a smooth strictly positive density, and every finite polynomial moment. These conclusions apply to the constructed tanh/sigmoid networks, including their frozen-output-bias ablation.

Proof. First verify the bounded-gradient hypothesis for the stated networks. Let $d_0, \dots, d_L$, with $d_L = 1$, be fixed layer widths, and write $W_\ell = w_\ell \tanh \Theta_\ell$, $b_\ell = \tanh v_\ell$, $w_\ell = 2/\sqrt{d_{\ell-1}}$. Finite inputs and targets give bounds X_*, Y_* . Hidden activations and their derivatives have absolute value at most one. Thus hidden outputs satisfy $A_\ell \leq 1$, with $A_0 = X_*$, and the linear output satisfies $|f_\theta(x_i)| \leq F = d_{L-1}w_L + 1$. Set $D_L = 1$ and $D_\ell = d_{\ell+1}w_{\ell+1}D_{\ell+1}$. Backpropagation and $|\tanh| \leq 1$ give, for every latent coordinate,

$$|\partial_{\Theta_{\ell,jk}} f_\theta| \leq D_\ell w_\ell A_{\ell-1}, \quad |\partial_{v_{\ell,j}} f_\theta| \leq D_\ell.$$

There are finitely many coordinates, so $\|\nabla f_\theta\| \leq J < \infty$ uniformly. Hence every individual and averaged squared-loss data gradient has norm at most $G = (F + Y_*)J$. The sampled objective therefore gives $b_B = -\eta g_B$, $K = \eta G$, and $\nu = \eta \sigma_{\text{eff}} > 0$ exactly.

Put $q = |a|$ and $V = 1 + \|\theta\|$. The triangle inequality gives $PV \leq qV + 1 - q + K + \nu \mathbb{E}\|Z\|$. On $\|\theta\| \leq R$, all conditional Gaussian means have norm at most $qR + K$. Consequently their mixture density, on a target ball $\|u\| \leq r$, is bounded below by

$$(2\pi\nu^2)^{-P/2} \exp[-(r + qR + K)^2/(2\nu^2)] > 0,$$

where P is the active dimension. This gives compact minorization, Lebesgue irreducibility, and aperiodicity. Localizing the displayed geometric drift to a sufficiently large ball, the Foster–Harris theorem gives uniqueness and weighted total-variation convergence (Hairer & Mattingly, 2011). For each $p > 0$, subadditivity ($p \leq 1$) or Young’s inequality ($p > 1$) gives $P\|\theta\|^p \leq q_p\|\theta\|^p + C_p$, with $q_p < 1$. Iteration from a finite start, convergence in law, and Portmanteau give the invariant p th moment. Invariance expresses its density as a mixture of translated Gaussians with covariance $\nu^2 I$; uniformly bounded Gaussian derivatives justify differentiation and imply smoothness and positivity. Freezing a bias removes its coordinate: the same argument holds on the remaining space, where the forcing is still full rank. $\square$

For the reported $\mu = .05$ and $\eta \in \{.5, 10\}$, the outer factors are .975 and .5. The proof establishes recurrence independently of any measured tangent-growth sign; it proves neither positive Lyapunov growth nor a density mode count.

F ADDITIONAL EMPIRICAL DETAILS AND CONTROLS

F.1 SMOOTH NONCONVEX AND MATCHED-NOISE CONTROLS

The existing nonconvex campaign supplies a smooth SGD construction with a proved invariant law and separately measured positive tangent growth. Its batch loss is

$$L_B(x) = \frac{h_B x^2}{2} - \xi_B x - \frac{0.92}{2} [x^2 - \log(1 + x^2)], \quad \xi_B \mid h_B \sim \mathcal{N}(0, \sigma^2 h_B / B),$$

where h_B averages B independent equiprobable values 0.95 and 1.05. Every sampled loss is coercive, but the population Hessian attains -0.035 . The exact update is

$$x^+ = [1 - \eta(h_B - 0.92)]x - \frac{0.92\eta x}{1 + x^2} + \eta\xi_B.$$

At $\eta = 8$, the outer coefficient lies in $[-0.04, 0.76]$ and the nonlinear remainder is bounded. Global drift and Gaussian minorization therefore give a unique attracting invariant law with every polynomial moment finite, independently of the numerical tangent calculation. The reference at $x = 0$ has log gain 1.94427 and harmonic gain 6.97714. Three completed FP64 configurations, each with 16 chains and one million updates, give mean moving log growth $+0.18959$ for $B = 1$, $\sigma = 0.001$. These finite-horizon measurements suggest positive long-time growth but do not prove the smooth model's infinite-time exponent; the positive stationary exponent of the piecewise control in the main text is established analytically.

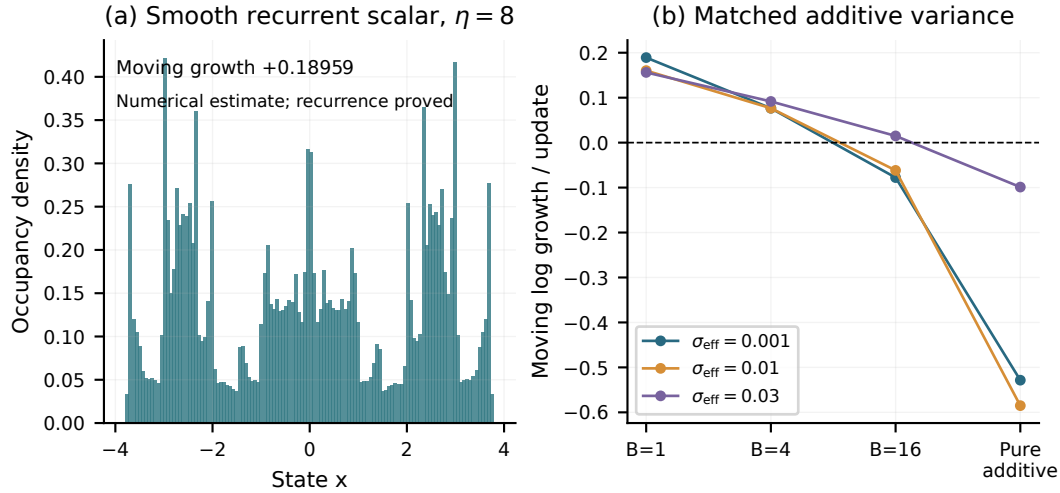


Figure 4: **Smooth recurrence and a controlled multiplicative-noise intervention.** (a) Pooled occupancy from all saved 20,000-update dense followups to the three million-update, 16-chain FP64 confirmations at $\eta = 8$, $B = 1$, $\sigma = 0.001$. The density is multimodal; no exactly-two-mode claim is made. The displayed moving growth is estimated from the primary long trajectories. (b) A separate intervention sets the per-example residual scale to $\sigma_{\text{eff}}\sqrt{B}$, so that $\text{Var}(\xi_B) = \sigma_{\text{eff}}^2$ stays fixed while $\text{Var}(h_B) = 0.0025/B$ changes. Points average completed million-update configurations; the pure-additive control sets $h_B = 1$. Thus batching can change the moving-growth sign even with matched unconditional additive variance.

The matched-variance results distinguish the two sources of randomness: at $\sigma_{\text{eff}} = 0.001$, moving growth changes from $+0.18917$ at $B = 1$ to -0.07775 at $B = 16$, whereas the pure-additive control gives -0.52845 . Conditional dependence between h_B and ξ_B remains as specified; matching variance does not make them independent. The finite-time means, correlated occupancy samples, and local density shape do not prove the sign of an asymptotic tangent exponent or a universal harmonic-to-bimodal transition.

F.2 CONSTRUCTED NEURAL RECURRENCE VERSUS ORDINARY-NETWORK CONTROLS

The constructed networks have two or four affine layers, hidden width four, and tanh or sigmoid activations, fitted to 64 fixed inputs sampled uniformly from $[-1, 1]^2$, with targets $\sin(3x_1) + 0.7x_1x_2 + 0.3\cos(4x_2)$. Effective weights and biases are bounded transforms of unconstrained latent parameters: $W = (2/\sqrt{\text{fan_in}}) \tanh \theta_W$, $b = \tanh \theta_b$. Their sampled objective is

$$\frac{1}{2B} \sum_{i \in I} (f_\theta(x_i) - y_i)^2 + \frac{\mu}{2} \|\theta\|^2 - \sigma_{\text{eff}} z^\top \theta, \quad z \sim \mathcal{N}(0, I).$$

Here the data gradient is globally bounded and the latent update has outer coefficient $1 - \eta\mu$; with $\mu = 0.05$ and $\eta \in \{0.5, 10\}$, recurrence with all moments follows from the global return argument. The full-rank parameter-linear forcing is an explicit design choice, distinct from regression label noise. Freezing the output bias restricts the state and forcing to the remaining active coordinates.

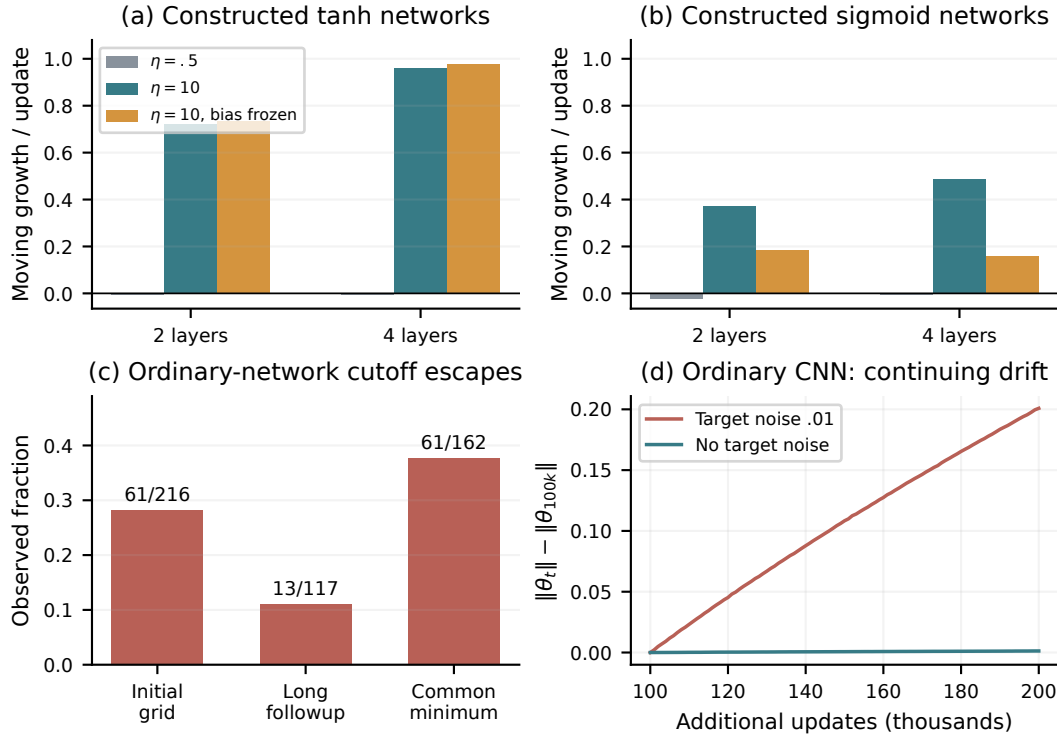


Figure 5: **Constructive positive controls do not imply ordinary-network equilibrium.** (a,b) At $B = 8$, $\sigma_{\text{eff}} = 0.003$, moving-growth means use the final 500,000 updates of completed million-update constructed-network confirmations, with 48 chains per setting. Large-step positive growth survives the frozen-output-bias ablation; four-layer tanh gives $+0.96022$ with trainable bias and $+0.97751$ with frozen bias. These are numerical growth estimates in models with separately proved recurrence. (c) The ordinary-network campaign retains all 495 completed cells, including 135 loss-cutoff escapes; cohorts have different designs and horizons, so these fractions are not a controlled comparison of escape probabilities. (d) A separate ordinary CIFAR-10 CNN at $\eta = 0.08$, $B = 128$ continues to drift over 200,000 additional updates. The displayed noisy and noiseless paths share their parent checkpoint and minibatch stream. A finite path’s norm drift does not measure ensemble-variance growth or prove mathematical escape.

In the matched CNN pair, the norm-mean shift between updates $[100000, 150000)$ and $[150000, 200000)$ is 0.10005 with target-noise standard deviation 0.01 , versus 0.000613 without added noise. Its tiny float32 moving-growth signs are unresolved and are not used to claim an invariant law. The broader ordinary controls include continuing parameter drift and dead-feature ReLU states whose remaining output bias follows a stable scalar recursion; these are not sustained positive-tangent neural equilibria. All figures use completed executions. The prospective next-study protocol contributes no results or counts.

F.3 NEURAL EXPERIMENT DEFINITIONS AND AUDIT

Training objective and sweep aggregation. The CNN and MLP use a fixed subset of 8,192 CIFAR-10 training examples. The objective is the example mean of the sum of squared errors over ten outputs; the optimizer is SGD without momentum, weight decay, clipping, or added target noise. The MLP has two hidden layers of width 512 with SiLU activations and 1,841,162 parameters; the CNN has three convolutional layers with SiLU activations. Figure 2 retains every batch size in $\{8, 32, 128, 512, 2048, 4096\}$ and every tested step size for the two architectures: $\{0.04, 0.08, 0.16\}$ for the CNN and $\{0.01, 0.02, 0.04\}$ for the MLP. Each cell averages three run-level estimates, computed after pooling that run’s last three available checkpoint probe sets. The checkpoint horizons vary across runs and cells. The reported harmonic statistic is therefore the mean of run-level harmonic statistics, not the harmonic statistic of a pooled mixture across runs. All 72 displayed means were recomputed from the immutable saved gains and agree with the published numerical audit to within 2×10^{-12} . Finite inverse-moment estimates do not establish finiteness of the population inverse moment; rare gains near zero can dominate that quantity.

Four diagnostic objects. For a fixed reference axis q , the signed scalar compression is $a_q = 1 - \eta q^\top H_B q / \|q\|^2$. Its moment estimates use $|a_q|$: $\widehat{M}_q(p) = \text{mean } |a_q|^p$, $\widehat{\Lambda}_q = \text{mean } \log |a_q|$, and $\widehat{H}_q = 1/\widehat{M}_q(-1)$. This differs from the reset-axis norm $\|(I - \eta H_B)q\|/\|q\|$, which retains the orthogonal part of the update. Both differ from a continuously propagated tangent vector $v_{t+1} = (I - \eta H_{B_t}(\theta_t))v_t$. The plotted cumulative tangent growth sums the log normalization factors while preserving the transported direction after every update. Three initializations are used: the leading Ritz vector, the full-data gradient, and a random direction. Multiple initializations are an empirical check and do not certify an asymptotic top exponent.

Continuation design and fixed frames. Four 4,096-update continuations start from retained reference weights: CNN discovery and confirmation streams at $B = 128$, $\eta = 0.08$ (reference update 20,000), and MLP random-reshuffling and iid-subset streams at $B = 4096$, $\eta = 0.04$ (reference update 10,001). These are fresh continuations, not bit-identical RNG resumes. Each iid minibatch is a uniform subset without replacement internally; batches are sampled independently across updates. The 8,192-example training set forms two batches per MLP reshuffling epoch, making the iid control essential. Four axes are held fixed: a full-data leading Ritz vector, the full-data gradient, and two random vectors orthogonalized against the Ritz vector. The MLP leading Ritz estimate uses 30 fully reorthogonalized Lanczos steps; its eigenpair residual is 2.80×10^{-7} relative to the Rayleigh value 68.1030. This certifies a small eigenpair residual, not global extremality. The reference state and all directions are reused by hash.

Window selection and shape checks. Every fixed coordinate is examined at widths 64, 128, 256, 512, and 1,024 updates, with 50% overlap. Raw coordinates are primary; detrending is diagnostic only. The screen requires exactly two prominent KDE maxima at bandwidth factors 0.7, 1, 1.4 times $n^{-1/5}$ times the window standard deviation, peak prominence at least 15% of the density maximum, separation at least 0.8 standard deviations, a valley below 80% of the smaller peak, and at least 10% of observations on either side. This is a descriptive screen, not a calibrated hypothesis test for dependent data. The displayed MLP interval is the first 512 updates of the late half, updates 2048–2559; it is an illustrative post-run display choice and also passes the broader bandwidth factor two. The same top-direction screen passes all 31 windows of width 256, all 15 windows of width 512, and all seven windows of width 1,024, under both MLP samplers. These overlapping counts are not independent replications. Fixed-gradient coordinates are more sensitive to window length; changing centers and amplitudes can merge lobes when a longer interval is pooled. Lag-one correlation and ordered traces remain essential: phase slips can weaken a global even/odd split while the process remains strongly alternating.

Paired-state probes and precision. At the initial state, 128 independent minibatches estimate the conditional compression laws. At each adjacent pair of updates $512k$ and $512k + 1$, $k = 0, \dots, 7$, the same 24 fresh minibatches probe both states; these probes do not alter training. For the iid MLP pair starting at update 2048, the fixed leading-axis harmonic gains are 1.17495 and 0.55872. Thus the observed lobes cannot be assigned one invariant harmonic regime along this axis. Training and Hessian products use float32 with TF32 disabled, with float64 reductions for coordinates, Rayleigh

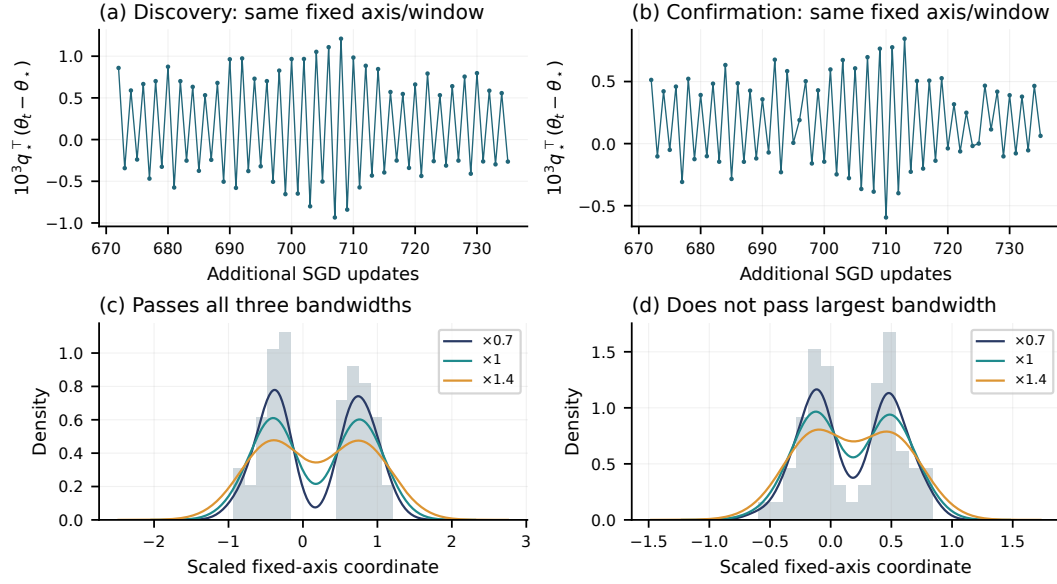


Figure 6: **A CNN short-window candidate does not confirm.** The discovery-only procedure selects the leading fixed axis and updates 672–735; the identical relative window is examined in a separate batch stream from the same parent checkpoint. Both time traces are shown alongside raw occupancy densities. The confirmation density has two prominent peaks at bandwidth factors 0.7 and 1, but only one at factor 1.4, failing the fixed screen. A visually suggestive short-window histogram alone is therefore insufficient evidence.

quotients, and norm ratios. Sparse same-state float64 derivative comparisons have a maximum MLP log-gain discrepancy of 1.77×10^{-7} ; this is a local derivative check rather than a global propagation error bound. An explicit small-network Hessian comparison and same-minibatch finite-difference check also pass.

F.4 AN ARCHITECTURE AND DATASET ATLAS

These seven pages retain the remaining distinct architecture and dataset views from the accepted experimental collection. We plot the two diagnostics relevant to the scalar edges, $\hat{\Lambda}$ and $\hat{H}$, over every saved step-size/batch-size cell. Run-level statistics use final-three-available-checkpoint pools before averaging, with varying checkpoint horizons for vision models. Gray cells marked N/A have no gain estimate; they are not stability observations. An E marks a recorded loss-cutoff escape, including any available pre-escape probes. Colors use a signed logarithmic transform about zero or one, with panel-specific ranges; numerical labels and colorbar ticks remain in the original units. A trailing + or – preserves the side of a reference after rounding. The archive provenance retains the original source figure, checkpoint windows, missing cells, and every displayed numerical value.

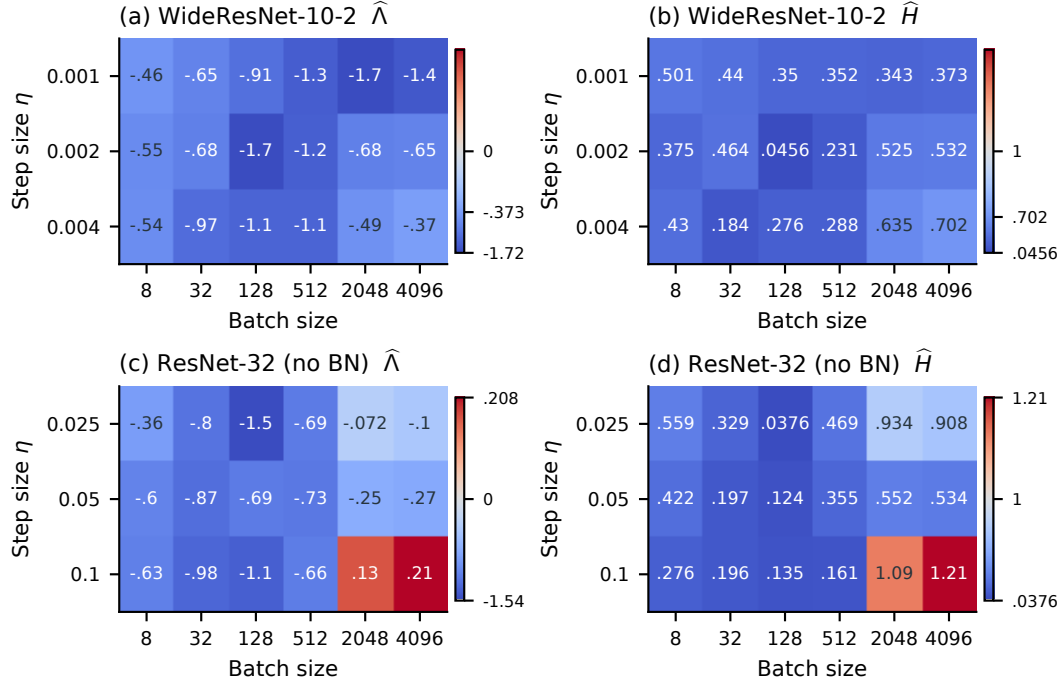


Figure 7: **New residual-network sweeps on CIFAR-10.** WideResNet-10-2 with native Fixup initialization and ResNet-32 without BatchNorm; plain SGD with squared loss on the fixed 8,192-example subset. WideResNet’s measured cells have negative scalar log gain and harmonic gain below one. ResNet-32 crosses both scalar references at its largest step size and large batches. Thus the main CNN/MLP comparison does not imply that all trained architectures exceed either reference. These are local scalar diagnostics, not network product exponents.

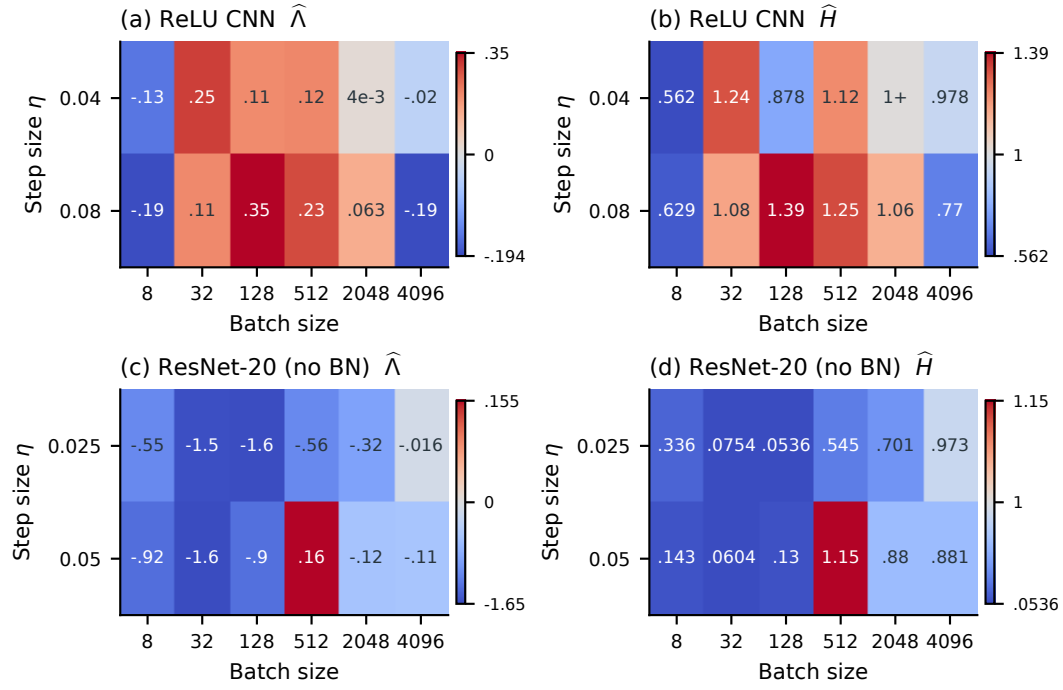


Figure 8: **Archived convolutional and residual architectures.** ReLU CNN with three convolutional layers and ResNet-20 without BatchNorm, using the same CIFAR-10 subset and squared-loss SGD objective. The CNN contains positive-log-gain cells on either side of the harmonic reference; for example $(\eta, B) = (0.04, 128)$ has $(\hat{\Lambda}, \hat{H}) \simeq (0.113, 0.878)$. ResNet-20 is mostly scalar-contracting in this grid, with a cell above both references at $(0.05, 512)$. These archived measurements complement the new sweep and retain their original checkpoint windows; they are not additional observations from the new continuations.

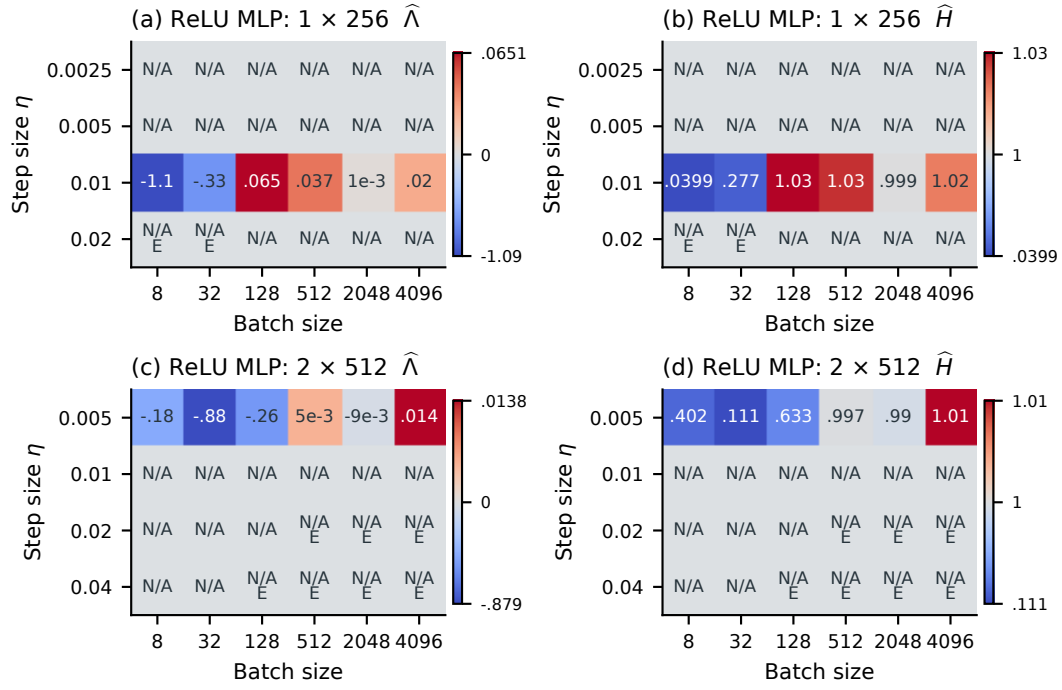


Figure 9: **Archived shallow ReLU MLPs, including incomplete diagnostic coverage.** One hidden layer of width 256 and two hidden layers of width 512; CIFAR-10 squared-loss SGD. Only one learning-rate row per architecture has gain estimates in the accepted archive; the other tested rows and recorded escapes remain visible as N/A and E. For the measured one-layer row, the large-batch cells approach the scalar references: at $(0.01, 128)$, $(\hat{A}, \hat{H}) \simeq (0.0651, 1.034)$. The missing rows do not support interpolating a stability boundary or assigning the architecture a universal regime.

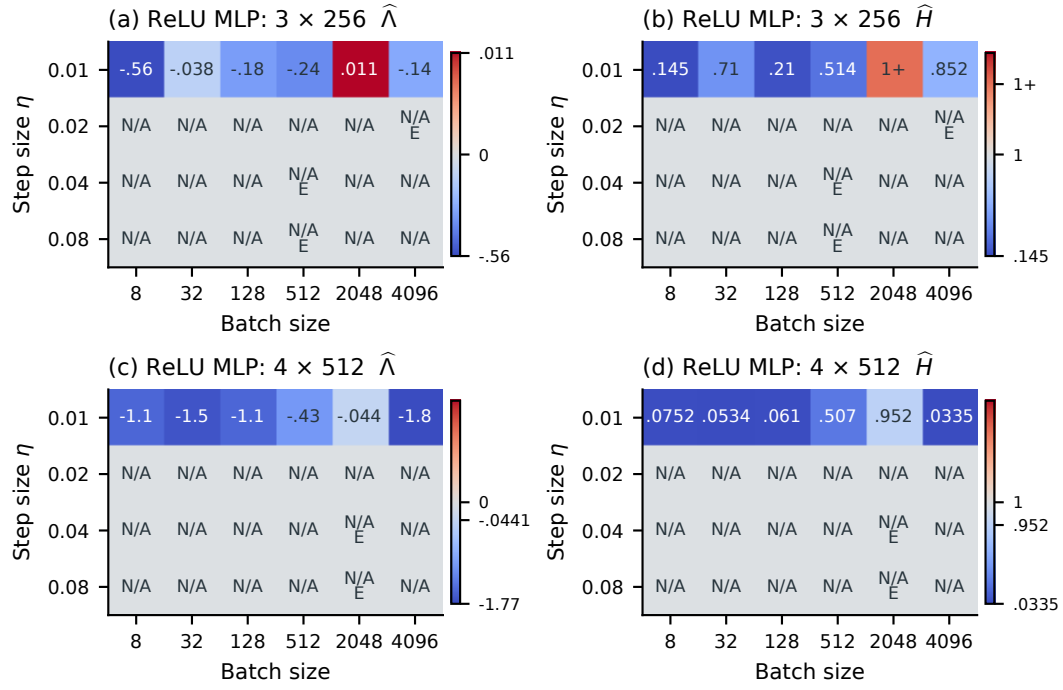


Figure 10: **Archived deeper ReLU MLPs.** Three hidden layers of width 256 and four hidden layers of width 512; CIFAR-10 squared-loss SGD. All available rows, missing gain estimates, and recorded escapes are retained. The measured three-layer row includes both contracting cells and a large-batch cell close to the two scalar references; the four-layer measured row is scalar-contracting. The depth comparison also changes hidden width and should not be interpreted as a controlled depth-only effect.

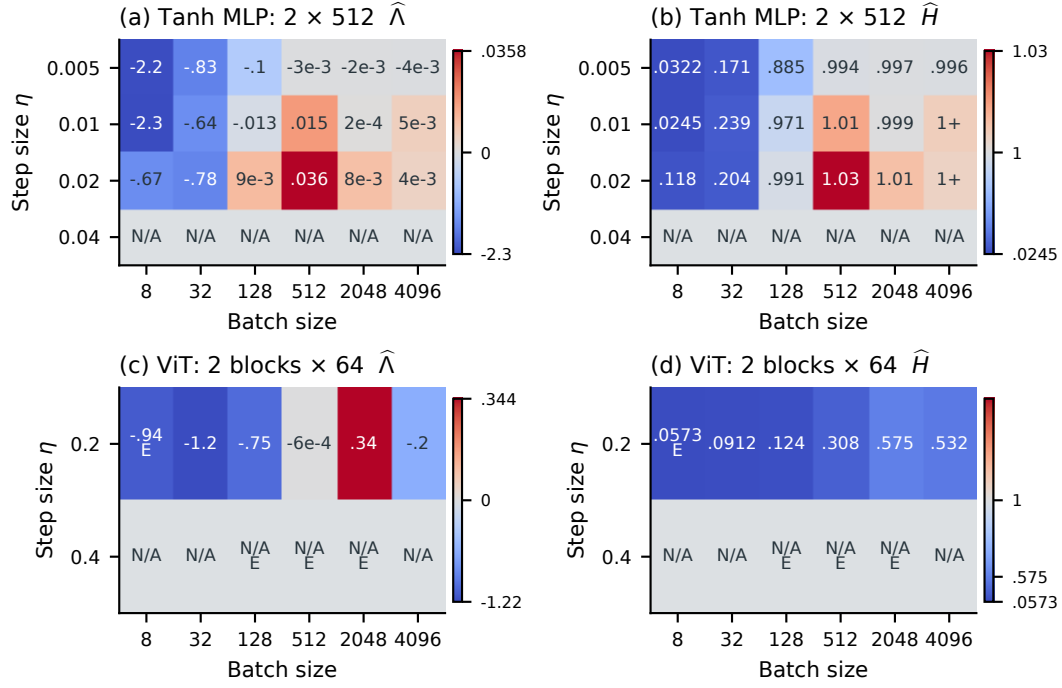


Figure 11: **Archived tanh MLP and small vision transformer.** Two hidden layers of width 512 with tanh activations, and a two-block width 64 ViT; CIFAR-10 squared-loss SGD. Several large-batch tanh settings approach the harmonic reference; at $(0.02, 512)$, $(\hat{\Lambda}, \hat{H}) \simeq (0.0358, 1.0328)$. The ViT at $(0.2, 512)$ instead has log gain near zero while harmonic gain is substantially below one: $(-0.000608, 0.308)$. Its $(0.2, 2048)$ checkpoint pool is log-expanding with $\hat{H} < 1$. These window-level differences illustrate distinct scalar regimes, while gray cells and E annotations retain the archive’s coverage and cutoff limitations.

Language-model metric definitions. The 168M-parameter decoder GPT is trained on a FineWeb-Edu subset using Adam and mean token cross-entropy. Each run processes 16.384 million tokens with context 256; the displayed pools combine checkpoints at 8.192, 12.288, and 16.384 million tokens. Let C_B be the Hessian or the positive-semidefinite generalized Gauss–Newton matrix (GGN). Raw curvature uses $s_B = g_B^\top C_B g_B / (g_B^\top g_B)$; the raw gain $1 - \eta s_B$ is an SGD counterfactual at an Adam-trained checkpoint. It is not an Adam stability multiplier.

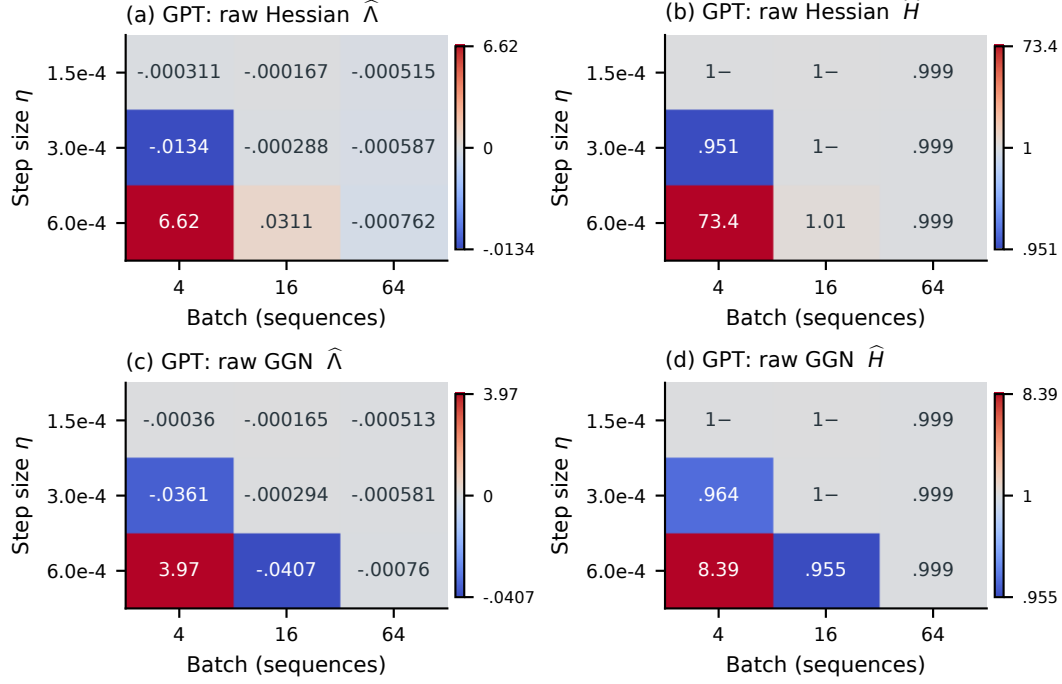


Figure 12: **GPT/FineWeb-Edu: raw Hessian and GGN scalar views.** The same nine parent hyperparameter cells underlie both views. Batch size counts 256-token sequences, and equal processed-token budgets imply different optimizer-update counts across batch sizes. The largest-step, smallest-batch corner is strongly expanding in both views; the exact Hessian and PSD GGN otherwise need not give the same scalar regime. Per-run pools contain only 48 Hessian or 24 GGN draws, so inverse moments remain exploratory. The extreme values are retained without clipping. These counterfactual scalar grids neither measure a full-Adam Jacobian nor establish a language model’s asymptotic stability boundary.

Frozen Adam geometry. Freeze the bias-corrected Adam second-moment preconditioner $D = \text{diag}[(\sqrt{\hat{v}} + \epsilon)^{-1}]$ at the checkpoint. The frozen-metric scalar curvature is

$$s_{D,B} = \frac{(Dg_B)^\top C_B(Dg_B)}{g_B^\top Dg_B}, \quad a_{D,B} = 1 - \eta s_{D,B}.$$

This is a symmetric-curvature Rayleigh quotient after a fixed change of coordinates. It omits both momentum dynamics and derivatives of the evolving preconditioner. The four GPT views are therefore distinct diagnostics of the same trained parents, not four architectures or four independent empirical replications.

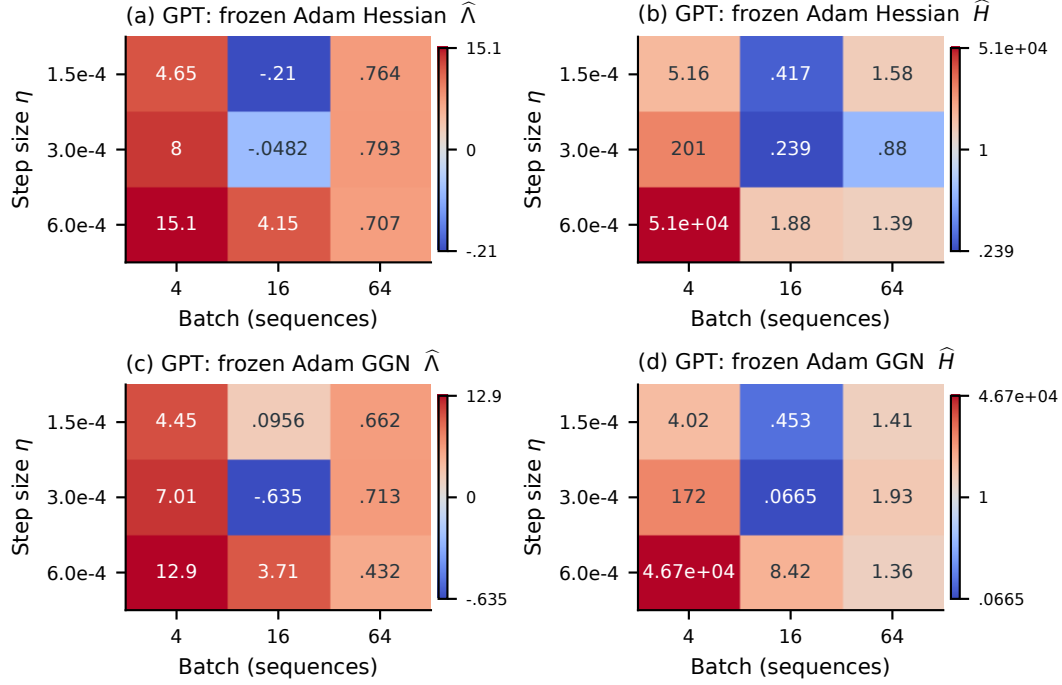


Figure 13: **GPT/FineWeb-Edu: frozen-Adam Hessian and GGN views.** Changing the frozen geometry changes the scalar regime map. For example, the frozen-GGN cell at $(\eta, B) = (0.0003, 16)$ has $\hat{\Lambda} \simeq -0.635$ and $\hat{H} \simeq 0.0665$, whereas other cells exceed both scalar references, some by large margins. All values use the same token-aligned pools and optimizer limitations as Figure 12. Local scalar expansion in these grids cannot be identified with instability of the full Adam state, and no stationary bimodal language-model law is claimed.